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Ban flower

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Declaration

This is to declare that the graduation project entitled (Ban flower) under the supervision of (Dr. Mohammed Maree) is our own work and does not contain any unacknowledged work or material previously published or written by another person, except where due reference is made in the text of the report.

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Abstract

In this project, we aim to develop a specialized web platform for a flower shop, allowing customers to browse available flower types, choose colors and arrangements, and easily customize a bouquet online. Users can view detailed information for each flower, including name, color, price, and image, with the ability to adjust the quantity of each type, enabling automatic calculation of the total price ,With 24-hour service for customers. The system offers an "AI Assistant" feature, where customers select the occasion (such as engagement, birthday, graduation...), and the system suggests the best flower types, colors, and bouquet designs based on a dedicated database. After completing the customization, the customer enters their information and selects a preferred pickup date, receiving instant notifications for order confirmation and updates. The platform also includes an authentication system to secure customer accounts, along with an admin dashboard to manage flower data and orders. Additionally , the flower shop provides a VIP seating area available for subscription, offering a cozy and aesthetic space for studying or relaxing. Subscribers also have the opportunity to showcase and promote their own small businesses in the shop's space, with support for product photography in a beautiful floral setting. This added service enhances customer engagement and provides extra value to the community through exclusive features and collaborative opportunities. The platform is expected to enhance the flower shopping experience, save customers time and effort, and support the flower shop in expanding its customer base and improving service quality through modern technology and smart recommendations.

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List of Abbreviations

AI: Artificial Intelligence

UI: User Interface

UX: User Experience

DB: Database

DFD: Data Flow Diagram

PK: Primary Key

FK: Foreign Key

Info: Information

ID: Identifier

OTP: One-Time Password

CRUD: Create, Read, Update, Delete

API: Application Programming
 Interface

CSS: Cascading Style Sheets

HTML: HyperText Markup Language

JS: JavaScript

JSON: JavaScript Object Notation

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Chapter One. Introduction

1.1 Background and Motivation

Today, people desire convenience and personalization in everything in life—gift-giving as well. Flowers have long been the traditional means of showing emotion and have been used for a range of occasions such as engagements, birthdays, and graduations. However, traditional flower shopping takes a lot of time and is not customizable. This project bridges this gap by developing a modern, AI-driven web platform for a flower shop where customers can browse, personalize, and order flower bouquets easily. The idea for this work stems from the growing need for digitalization in small businesses and the potential to improve customer experience using intelligent recommendation engines and real-time interaction.

1.2 Aims and Objectives

The overall object of our proposed system is to develop a user-friendly, AI-enabled web platform by which customers can design and order custom flower bouquets online. In particular, we aim to:

1. Develop an interactive catalog of flower types with data (name, color, price, image).
2. Provide functionality for users to select flowers, and adjust quantities with price updates.
3. Develop an "AI Assistant" that provides bouquet design suggestions based on selected occasions.
4. To deliver order confirmation and real-time update notifications to customers.
5. Offer a subscription feature for VIP seating inside the flower shop, where users can study and relax or promote their small projects, with support for product photography.
6. Providing 24-hour service to customers.

1.3 Problem Statement

Based on a survey that we conducted to assess the need for the proposed platform (please refer to [Annex 1](#)), it was clear that many customers are finding it difficult to select and customize flower arrangements due to time limitations in-store, ignorance of flowers, or absence of guided assistance. Accordingly, we attempt to address the following issues by developing our proposed system:

- Designing an intelligent online platform to simplify the process of shopping flowers with meaningful customization and AI recommendations.
- Offering dynamic bouquets for specific events.
- Enabling users to manage flower stock and track their orders through a web-based admin interface.

1.4 Contributions

We summarize the main contributions of our proposed work as follows:

- Design and development of a custom flower shopping experience using the latest web technologies.
- Implementation of a recommendation engine powered by AI on the basis of occasion-specific preferences and ratings.
- Real-time notification system for order updates to customers.
- An admin-friendly interface for easy flower and customer order management.
- A responsive design ensuring ease of access through different devices.
- Integrating a VIP seat subscription feature that allows users to study, relax, or showcase their own work
- Providing 24-hour service to customers.

1.5 Structure of the Report

This report is organized in the following chapters:

Chapter One: Introduction – Background, motivation, problem statement, objectives, contributions, and the general structure.

Chapter Two: Overview, Existing Systems, Comparative Analysis, Summary.

Chapter Three: Provides the system analysis, including an introduction to how the system works, functional and non-functional requirements, Data Flow Diagram (DFD), Use Case Diagram, Use Case Descriptions, and Sequence Diagrams.

Chapter Four: Describes the system design, such as the Class Diagram, Database Design, and Interface Design.

Chapter Five: Presents the conclusion and outlines possible future enhancements and features that could be added later.

Chapter Two. Literature Review

2.1 Overview

This chapter presents a comprehensive review of the existing systems and technology with respect to online flower shopping via web-based platforms with focus on certain features that are well-suited to this project's objectives. This highlights current market solutions, functionalities behind them, as well as limitations, particularly of personalization, user interface, and usage of artificial intelligence. Specific interest is given to the use of artificial intelligence in assisting users in creating customized bouquets according to occasion, and how intelligent recommendation systems influence business growth and customer satisfaction. Moreover, the chapter explores typical system architectures, user interface patterns, and management tools in similar platforms. The discussion sets the context for how gaps in available solutions are established and justifies the proposed system's value proposition—i.e., a user-friendly, intelligent web platform for flower shops by design to streamline customer experience and operations.

2.2 Existing Systems

In this section, we review and analyze the most relevant online markets focused on flower sales. We present a brief about each reviewed system, its main features with a screenshot of its main page.

1. [Floward \[1\]](#)

While Floward excels at offering finished, occasion-specific bouquets through a neat interface and precise tracking, our proposed platform takes personalization to another level by enabling customers to personally choose every variety of flower

separately, define exact quantities, and shape every arrangement petal by petal. An in-built AI assistant will throw color scheme and idea of design based on the event selected, but users can always overrule ideas or build entirely from scratch. Unlike Floward's region-locked presets, our platform combines this deep customization with AMP-supporting pickup scheduling, robust authentication flow, and a bespoke admin dashboard—empowering smaller stores to manage inventory, orders, and AI-powered offerings in underpenetrated regions. **Strengths:** User-friendly interface with modern design ,Offers gifts and flowers, increasing options ,Effective delivery tracking system **Weaknesses:** Limited availability in small towns like Jenin ,Some bouquets only available within certain regions

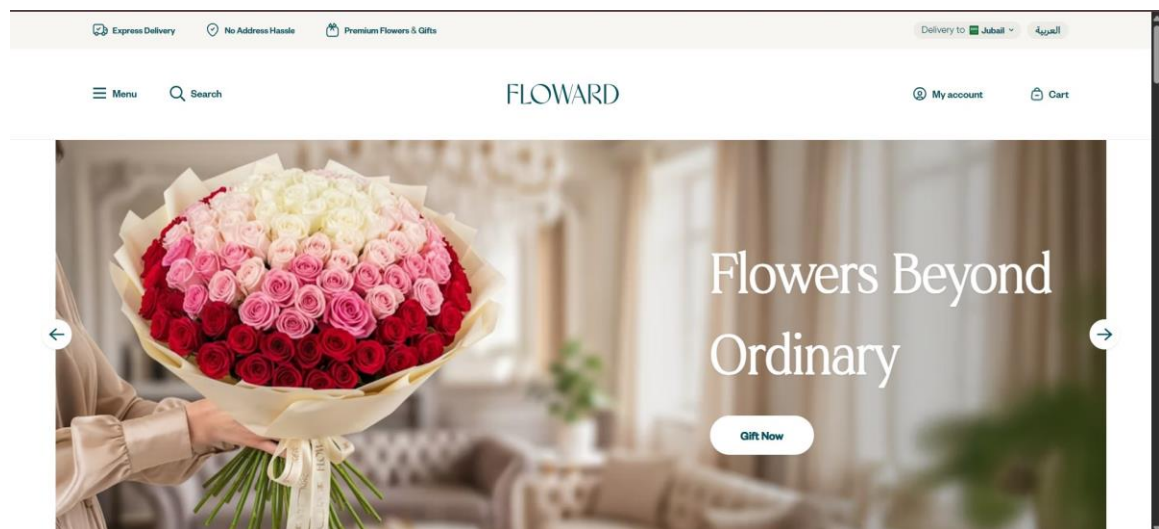


Fig 1. Screenshot of the main page of [Floward \[1\]](#).

2. [Ferns N Petals \(FNP\) \[2\]](#)

Ferns N Petals has a mammoth range of same-day and midnight deliveries and personalized gift suggestion, its messy UI and premium overseas shipping fees intimidate fresh shoppers. While our competitors complicate things, our website streamlines the process with users browsing through a chic bouquet-builder—allowing them to manually choose each bloom, define exact amounts, and enjoy AI-enabled design suggestions for any occasion. By our focus on local pickup scheduling, real-time notifications, and simple but secure authentication flow, we eliminate high shipping charges and interface cruft, and our bespoke admin dashboard gives smaller stores in under-served neighborhoods full control over inventory, orders, and AI-analyzed offerings. **Strengths:** Large variety of flowers and arrangements ,Fast logistics and delivery ,Personalized

salutations and gift recommendations**Weaknesses:** Complicated interface for new customers ,Hefty overseas shipping fees

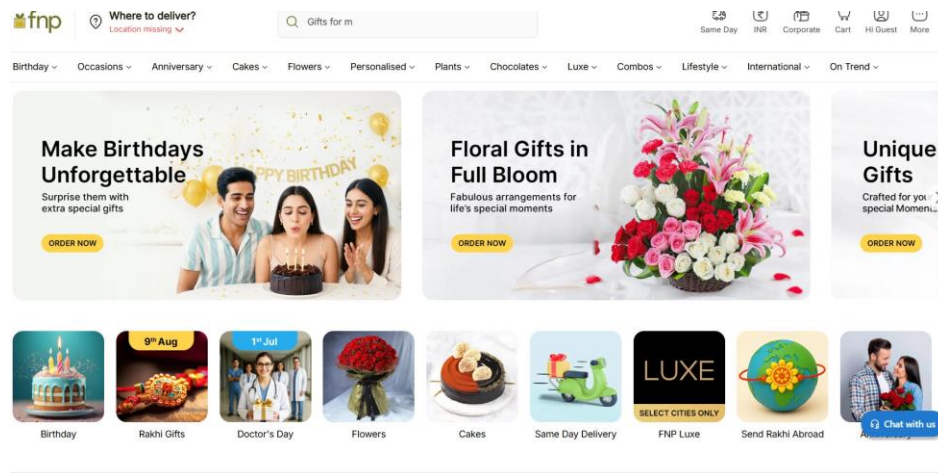


Fig 2. Screenshot of main page of [Ferns N Petals \(FNP\) \[2\]](#).

3. [Interflora \[3\]](#)

Interflora is dependent upon an enormous network of local florists to professionally customize same-day international delivery, its upscale prices and limited bouquet personalization have shoppers craving greater freedom of choice. Our site, conversely, invites users to personally choose their own flower species, colors, and quantities—and then edit each arrangement petal by petal or go for AI-generated design recommendations for any celebration. With a focus on adaptive local pickup scheduling, real-time order alerts, and a simple, uncluttered interface with robust authentication, we cut out exorbitant shipping fees and interface overhead. A customized admin dashboard allows small stores in under-served markets to track inventory, orders, and AI-driven offerings, delivering a well-personalized flower-purchasing experience.**Strengths:**International reach ,Professionally designed arrangements by florists ,Same-day delivery option**Weaknesses:**Fairly expensive ,Limited direct personalization

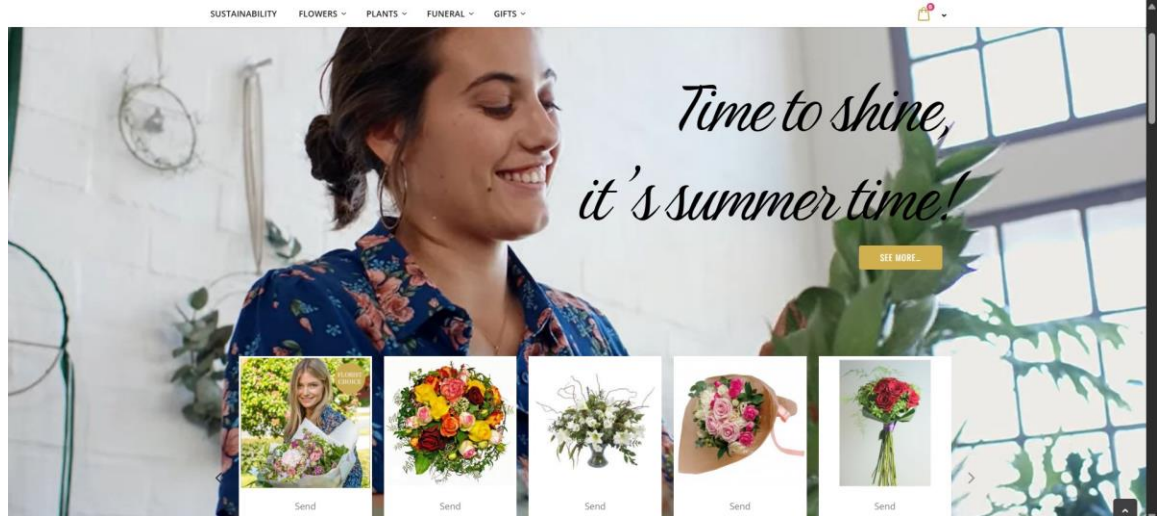


Fig 3. Screenshot of main page of [Interflora \[3\]](#).

4. [1-800-Flowers \[4\]](#)

1-800-Flowers leads the pack in excellent customer service, overwhelming numbers of bouquets to select from, and robust mobile visibility, its presence is usually limited to the U.S. (and a handful of other nations), and pricey high-end costs deter thrifty consumers. In contrast, our website invites consumers everywhere we're available—region, regardless—to design absolutely custom arrangements by selecting and combining every variety of flower, color, and quantity, from scratch or from AI-generated design inspiration. By focusing on local pickup (eliminating extravagant shipping fees), a clean, user-friendly design, live order alerts, and a robust authentication cycle, we give independent stores absolute control over inventory and orders through a customized admin dashboard—delivering unique creativity without the cost or sophistication of large-chain logistics.

Strengths: Superb customer service ,Great bouquet selection ,Large mobile app

Weaknesses: Only in the U.S. and a few countries ,Relatively expensive

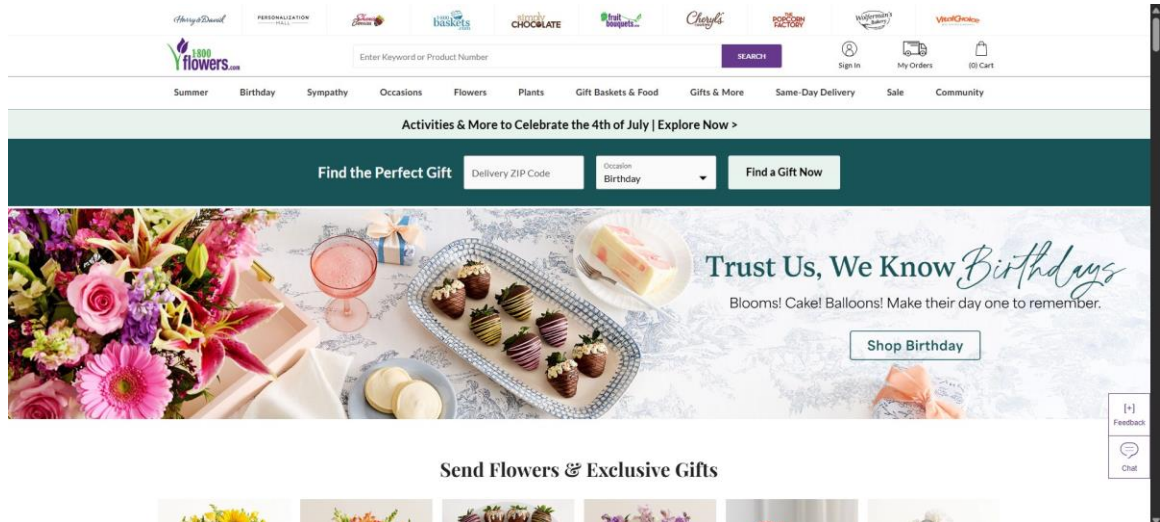


Fig 4. Screenshot of home page of [1-800-Flowers \[4\]](#).

5. [ProFlowers \[5\]](#)

ProFlowers excels at delivering low-priced, farm-fresh bouquets with seasonal promotions, but its customization is limited and delivery is only to certain regions. On the other hand, our platform enables customers to personally choose each bloom's color, type, and number—or leave custom design to an AI assistant—then select a local pickup date to keep shipping within limits. Underpinned by instant notifications, secure login system, and customized admin dashboard to track inventory and orders, we enable true creative freedom and control of operations for small stores in under-served markets. **Strengths:** Low cost ,Guarantee of freshness ,Seasonal promotion ,**Weaknesses:** Geographical restrictions ,Fewer options to personalize

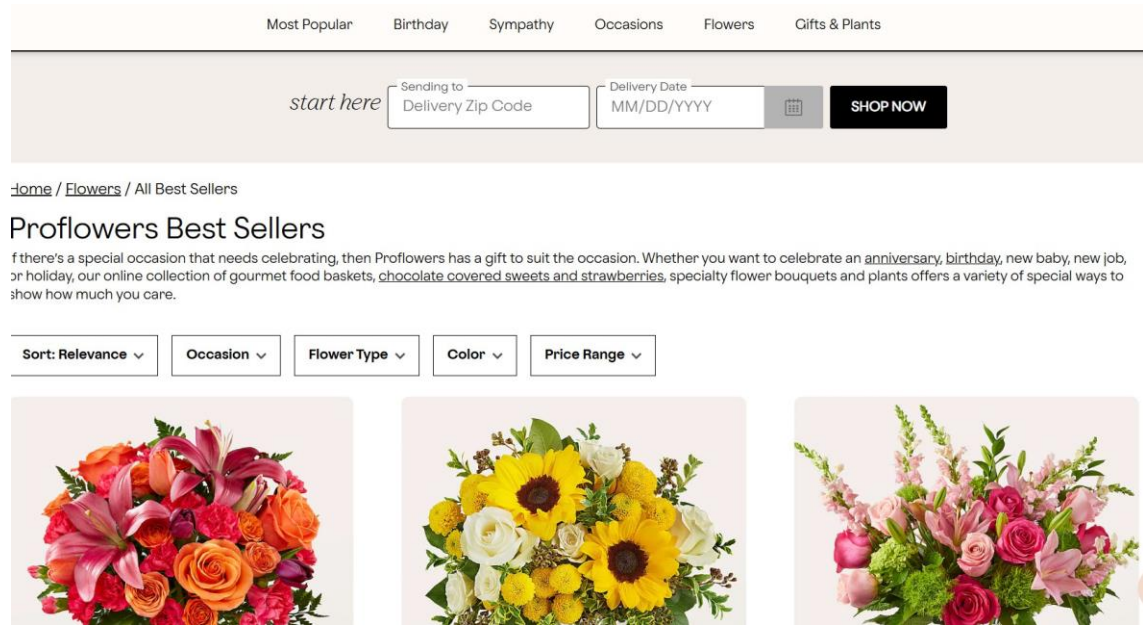


Fig 5. Screenshot of home page of [ProFlowers \[5\]](#).

6. [FloraQueen \[6\]](#)

FloraQueen offers clean, multilingual web interfaces and global shipping to over 100 countries but limited real-time bouquet design and sluggish customer support, our website enables users to personally select each flower type, color, and quantity—or have an AI assistant do occasion-based designs—and subsequently choose flexible local pickup to sidestep shipping constraints. On the basis of a clean, user-friendly design with immediate order notifications, robust authentication, and a tailored admin dashboard, we enable greater personalization and operation control for flower shops from any geography. **Strengths:** International shipping ,Modern and clean design ,Support for multiple languages .**Weaknesses:** Customer care response time is too slow ,Less real-time customization

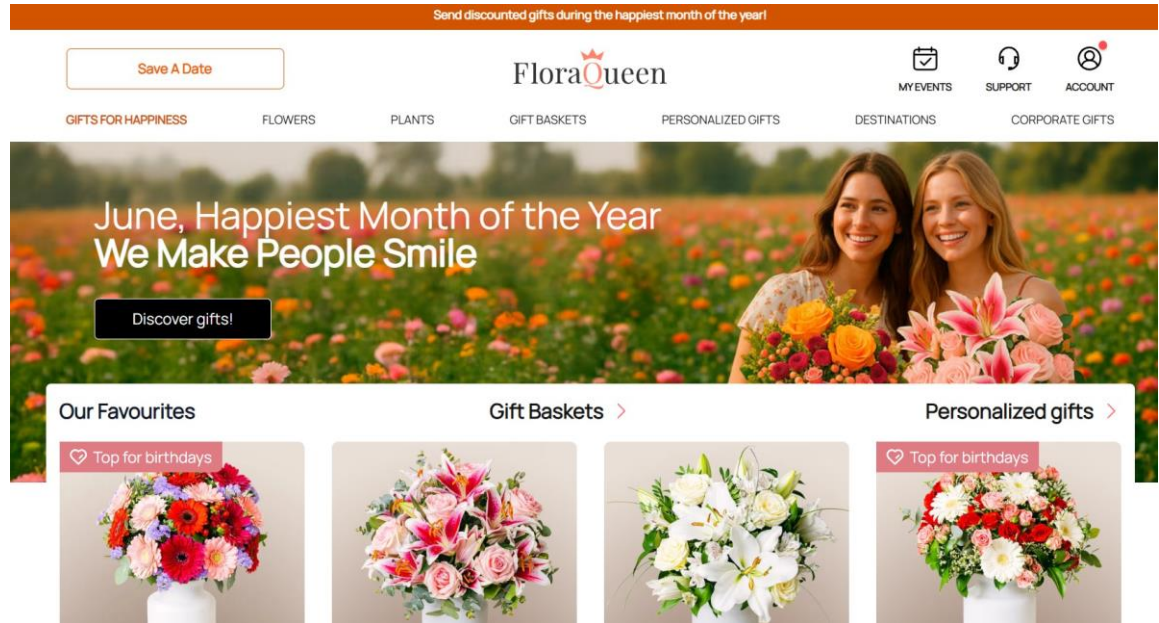


Fig 6. Screenshot of [FloraQueen \[6\]](#) main page.

7. [Zhoor Alreef \[7\]](#)

Zhoor Alreef shines with elegant product displays, luxury gift sets, and reliable local delivery within Saudi Arabia, its flower offerings are ancillary and not elsewhere available outside that market. Our website is built on flowers first—allowing shoppers to navigate through all flower types, blend colors and quantities as preferred, and receive recommendations from an AI assistant for personalized bouquet construction. We combine that with flexible pick-up scheduling, in-order notifications, secure user authentication, and a customized admin console—allowing flower stores in any region to manage inventory, orders, and custom offerings where the big boys haven't moved in yet.

Strengths: Advanced presentation of products ,Luxury gifts ,Local delivery guarantee

Weaknesses: No emphasis on flowers ,Not available outside Saudi Arabia

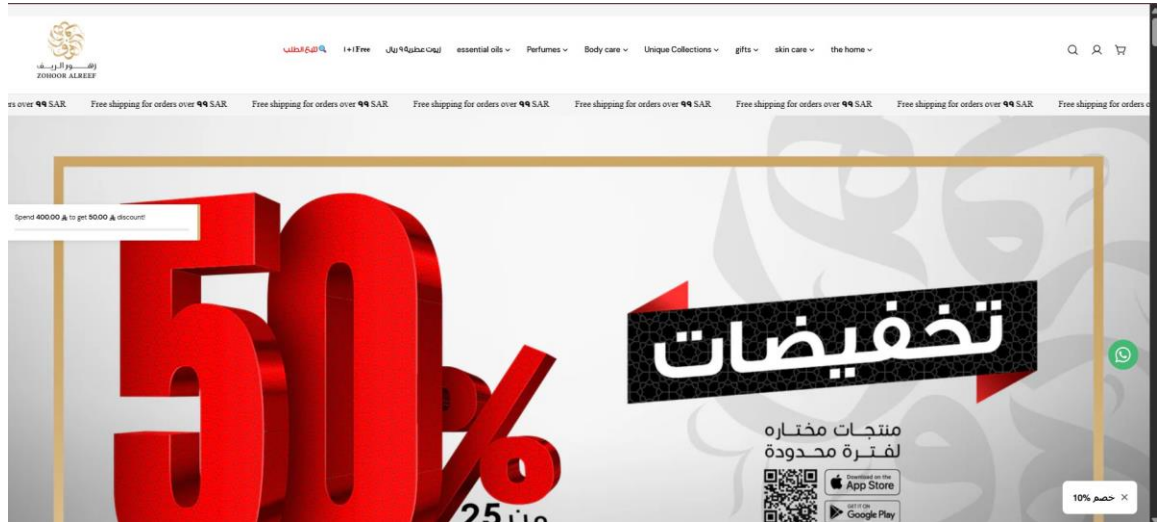


Fig 7. Screenshot of [Zhoor Alreef \[7\]](#) main page.

8. [Blommarket \[8\]](#)

Blommarket enjoys the benefit of outfitting neighborhood florists with same-day delivery at reasonable prices, its austere presentation and Saudi-only accessibility might be restrictive. In contrast, our site positions flowers center stage—enabling customers to personally choose each bloom type, hue, and number or leave to AI-powered design suggestions—then choose flexible pickup dates to circumvent shipping limitations. Backed by instant reminders, secure login, and an enterprise-grade admin dashboard, we offer a richer, more personalized experience that small shops are able to offer anywhere they sell. **Strengths:** Saves local small stores, Efficient delivery, Reasonable prices. **Weaknesses:** Present only in Saudi Arabia, Minimal user interface

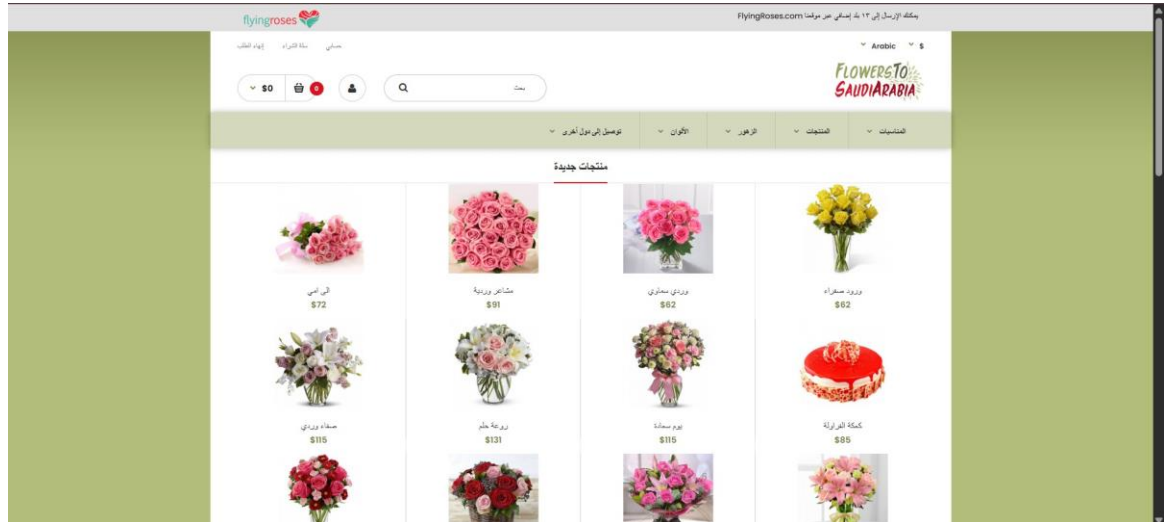


Fig 8. Screenshot of [Blommarket \[8\]](#) main page.

9. [Talabat Flowers \[9\]](#)

Talabat Flowers benefits from Talabat's vast network for express delivery and a recognized app interface—offering swift, reliable service in Gulf city centers—it only offers ready-to-pick bouquets with little customization leeway and is not yet available in smaller towns. In contrast, our platform is about total DIY autonomy: customers can individually choose each blossom, color, and quantity or trust AI-inspired design suggestions, then choose convenient local pickup dates, all from an easy-to-use, logical interface backed up by solid security authentication and a tailored admin control panel that allows smaller shops to serve any market they desire. **Strengths:** Having Talabat's infrastructure as a support ,Efficient and reliable delivery ,Familiar app environment for customers . **Weaknesses:** Available only in large cities ,Limited customization of bouquets

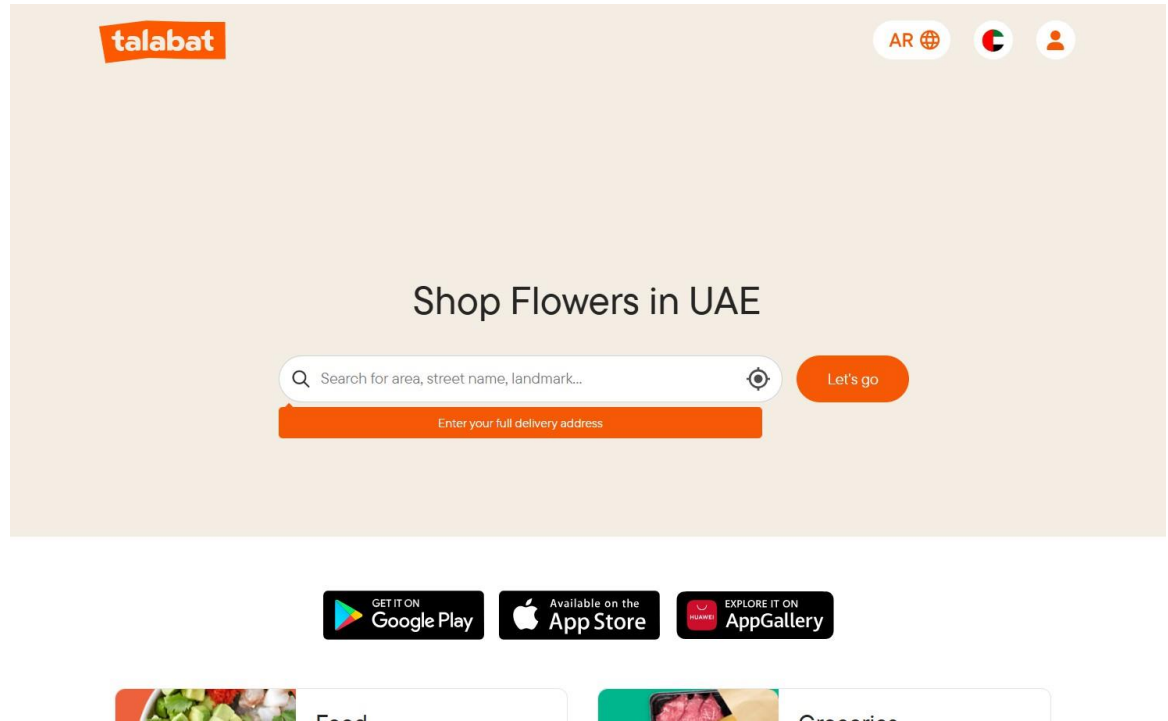


Fig 9. Main page screenshot of [Talabat Flowers \[9\]](#).

10. [Rose N' More \[10\]](#)

While Rose N' More dazzles with high-end luxury arrangements, master photography, and hand-picked bouquets for Gulf customers, its premium price tag and Gulf-exclusive shipping can price out potential customers. Our platform, on the other hand, is conceived for worldwide accessibility and creative liberty: customers choose each flower, color, and quantity or let AI-facilitated design cues guide any holiday, then choose convenient local pickup times to escape shipping restrictions. A crisp, user-friendly design safeguarded by robust authentication—and supplemented with a tailored admin dashboard to monitor inventory, orders, and AI recommendations—allows flower shops worldwide to display clearly distinct bouquets at any scale. **Strengths:** Luxury flower arrangements ,Professional product pictures ,Variety of bouquet types. **Weaknesses:** Too costly ,They are not found in non-Gulf nations.

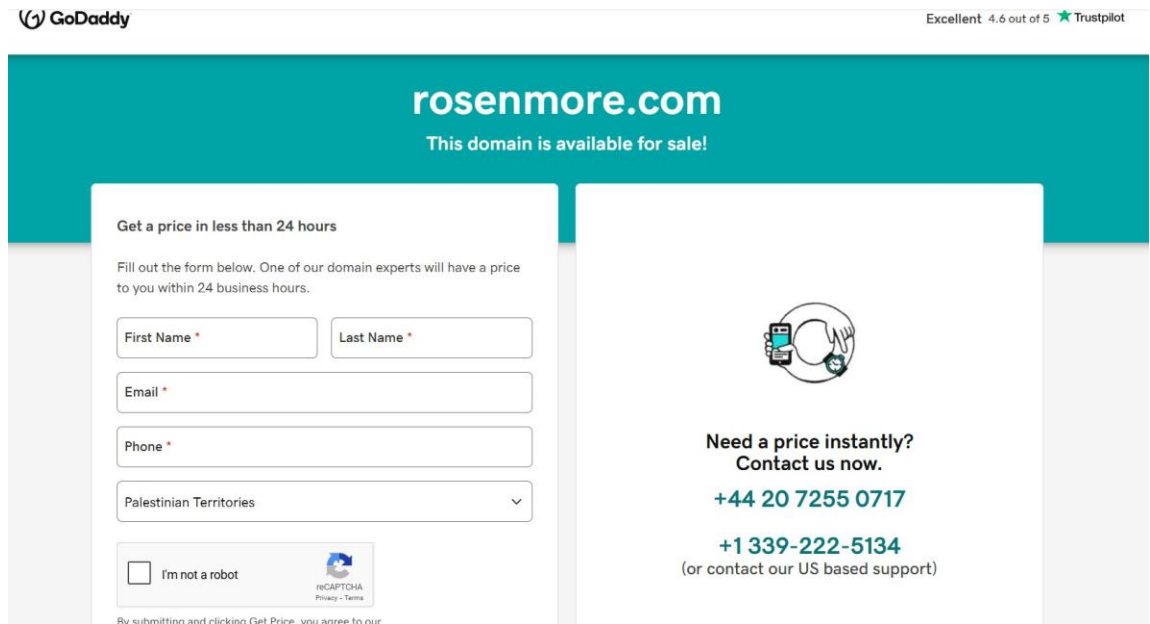


Fig 10. Screenshot of [Rose N' More \[10\]](#) main page.

2.3 Comparative Analysis:

We further compare the studies systems in the previous section according to the below comparison criteria.

guest:

He, in turn, goes to the site and browses the site from outside, looking at the products, but he can only shop by logging in to his account.

registered user:

He purchases from the website and selects the product he chooses by registering on the site.

Flower shop own:

He is the owner of the shop that provides products to users, meets their needs, and designs customer requirements as specified.

admin:

Add new flower types, modify or delete existing flowers, view all orders placed by customers, change order status (in progress - ready - delivered...), view the list of registered users, follow VIP subscriptions, accept or reject subscription requests.

1. Services:

Most contemporary sites merely offer standard services by presenting pre-designed flower bouquets to purchase or order, with some amount of customization in between. For instance, websites like 1-800-Flowers and ProFlowers offer users with pre-determined bouquet lists to select from, with hardly any personalization in regard to delivery date and location.

On the other hand, our system offers dynamic bouquet personalization, allowing customers to select flower types, colors, quantities, and calculate the total cost automatically. We also offer an "AI Assistant" that recommends the best bouquet design based on the selected occasion, making the platform smarter and more interactive than traditional solutions.

2. Target Audience:

Global platforms principally serve the overall population looking for flower delivery options, with certain of them covering corporate customers as well (e.g., flower subscription on a weekly basis to offices). Our platform serves the local market and spans various segments of users: individuals, event planners, and users organizing special events. The platform also has a customized look in Arabic and is tailored according to the needs and local environment.

3. Supported Language

The majority of existing platforms like Teleflora and FTD have the default language of English, and most only include limited translation capability. Our platform stands out as having native-level support for Arabic, with the Arabic speakers experiencing a smooth, easy-to-use interface, creating higher engagement and trust.

4. Targeted Region

Most current platforms aim at the world market, with a major focus on the U.S. and European markets. Some focus on specific countries or regions. Our platform is specifically tailored to the Palestinian market, offering a geographically and culturally tailored solution that serves the local audience's unique needs.

5. Content and Media

Others utilize mostly pictures to show floral arrangements, and some even include brief tutorial videos. Our site integrates high-quality pictures, complete descriptions, and an AI-based recommendation system on a proprietary database, resulting in a rich and interactive experience for users.

6. Goals:

The main aim of current systems is to facilitate the flower purchase and delivery process. Our system goes a step further by enabling customers to design the perfect bouquet, receive intelligent suggestions, and interact easily with the store, offering an encompassing and intelligent flower shopping experience.

Website System	Services	Target Audience	Supported Language	Target Region	Content & Media	Goals
<u>1-800-Flowers</u>	Ready-made bouquets, limited customization	Individuals	English	Global	Images, brief text	Easy flower ordering and delivery
<u>ProFlowers</u>	Occasion-based bouquets, seasonal offers	Individuals, businesses	English	USA	Images + basic descriptions	Fast delivery and pre-designed bouquets
<u>Interflora</u>	International flower delivery, corporate services	Individuals, companies	English, some EU languages	International	Images, short descriptions	Fast and convenient gifting
<u>Floward</u>	Bouquet selection, delivery, add-on gifts	Individuals, gift buyers	English, Arabic	Gulf region	Images, some videos	Gift customization and express delivery
Our Proposed Platform	Custom bouquet builder, AI assistant based on occasion, real-time pricing	Individuals, event organizers, local customers	Arabic, English	Palestine	High-quality images, articles, smart suggestions	Personalized and intelligent shopping experience

Table 1. Comparison between the reviewed systems

2.4 Summary

We have analyzed some of the current flower shop platforms and compared them with our proposed system in this chapter, highlighting the main differences between them. The comparison was based on various criteria like the services they offer, their target market, languages supported, geographical scope, display of content and media, and overall purposes.

Although most of the tested platforms provide simple functionalities like browsing and ordering pre-arranged flowers, they usually don't support detailed customization, smart suggestions, and local market awareness. Our system is designed to close these gaps by delivering a sophisticated web-based system that is adjusted to the requirements of Arabic-language speakers in Palestine.

Among the most significant aspects of our project is the incorporation of intelligent elements such as the AI Assistant that suggests suitable bouquets depending on the selected occasion. Secondly, the system has complete bouquet customization, rich content display, and shop-customer communication as top priorities. The new way offers a modern, user-centric, and culturally appropriate way of shopping that increases customer satisfaction as well as the quality of services.

Chapter Three: System Analysis

2.1 Introduction

In this chapter, we analyze the proposed flower shop web platform to identify and understand the existing limitations in traditional flower shopping experiences, and develop system requirements to resolve these problems with smart digital solutions.

This stage entails the identification of the requirements of the system, understanding the user roles, and modeling system behavior and data flow using tools such as UML diagrams, Data Flow Diagrams (DFD), Use Case Descriptions, and Sequence Diagrams.

The purpose is to ensure the platform is meeting the functional requirements of customers as well as administrators efficiently and securely.

2.2 Analysis of Existing Systems

We analyze existing traditional and online flower shopping habits here in order to identify current drawbacks and present rationale for a better and intelligent way.

To gather data and get a better picture of the present, a number of analysis techniques were used:

- Document Review:

In order to develop a competitive and robust flower shop web application, a careful review of documents was conducted. In this phase, the objective was to study existing solutions in the e-commerce of flowers in order to establish strengths, weaknesses, and best practices. The following documents were examined:

Online Flower Shop Websites : Large flower delivery services such as 1-800-Flowers, Interflora, and Teleflora were reviewed. These websites offered insight into user interface, flower categorization, level of customization, and payment processing. They excelled at simple navigation by the user and delivery scheduling, but fell short in intelligent assistant function and real-time personalization by occasion or intent.

Mobile Applications : Floward and FloraQueen apps were analyzed for mobile UI/UX, bouquet preview, and notification system. Their limitation was the inflexibility in creating bouquets as per personal choice.

Product Catalogs and Brochures : Online brochures of local flower shops were analyzed to understand trending flower combinations, pricing, and seasonal flowers. This guided the design of the database schema for flower attributes and pricing rules.

System Documentation of Open-Source E-Commerce Tools : PrestaShop and Magento were reviewed for architectural reference, with a focus on modules for real-time pricing calculations, customer profiles, and inventory management.

Through this review, a central gap in current flower platforms was identified: the absence of AI-driven personalization and dynamic bouquet customization. This project aims to fill the gap by introducing an AI Assistant, a modern bouquet design interface, and a real-time pricing engine—without sacrificing a smooth and secure customer experience.

- Data Collection:

Data were collected on seasonal trends, user preference, pattern of bouquet customization, and common problems in customer services to help in requirement development.

- User Interviews:

Interviews with customers and employees of flower shops were conducted to know their experiences, problems, and suggestions for improvement.

- Questionnaires:

Structured questionnaires were also distributed to a sample of potential users to establish what the most important aspects of an online florist would be, for example, browsing capabilities, AI recommendations, and delivery.

For more information, please refer to [Annex 1](#)

- Observation:

Behavior among customers in physical flower shops was monitored regarding how they handle flowers, staff, and arrangement options to inform the platform design.

- Sampling, and Research:

A representative sample of top flower shop platforms was compared to industry benchmark functionality, user experience, and technologies deployed in the backend.

2.3 System Requirements

This defines the functional and non-functional requirements of the flower shop's web platform. They are crucial to the user needs delivery and smooth and smart digital shopping experience.

2.3.1 Functional Requirements

Browse Flower Details

FR-1 :The system should display details of flowers including name, image, color, and price.

FR-2 :The system should allow users to browse through all types and categories of available flowers.

AI-Based Recommendations

FR-1 The system shall enable customers to select an occasion.

FR-2 The system shall display AI-recommended bouquets based on the occasion.

Customize Bouquet

FR-1: The system shall enable customers to select flower types, colors, and quantities.

FR-2 The system shall adjust the bouquet price automatically based on selections.

User Profile Management

FR-1 : The system must allow users to create accounts and safe log in.

FR-2 :The system must allow users to update their profile details (e.g., password, name, address).

Order Placement and Notifications

FR-1 : The system must allow users to order flowers with a chosen pickup or delivery date.

FR-2The system must alert users of order submission and status.

FR-3The system must alert the admin with order details.

Admin Management Features

FR-1: The system must provide the capability to the admin to manage flower stock (add/edit/delete).

FR-2 The system must provide the capability to the admin to view and manage all orders made.

FR- :3 The system must provide the capability to the admin to manage users and edit content.

VIP Seating Subscription Features

FR-1: The system should allow registered users to view information about the VIP seating subscription (features, pricing, usage terms).

FR-2: The system should allow users to subscribe to the VIP seating service through their profiles.

FR-3: The system should allow VIP users to submit business promotion requests (e.g., showcasing products).

FR-4: The admin must be able to manage VIP subscriptions, approve or reject bookings, and control promotional content visibility.

2.3.2 Non Functional Requirements

Usability:

The system must offer a user-friendly graphical interface.

The interface must be consistent for all pages with similar design elements (icons, toolbars).

Accessibility:

The system shall support accessibility features for individuals with disabilities.

The system shall support multilingual functionality (e.g., English, Arabic).

Reliability and Availability:

The system shall be designed to minimize downtime and faults using best practices.

Performance:

The platform shall be hosted on a reliable web server.

Performance may be hardware and internet connection dependent for the client.

Security:

Users should select safe passwords with requirements for complexity (e.g., numbers, uppercase letters).

The system should save user information securely and adhere to best practices for authentication.

2.4 Design Requirements

2.4.1 User Requirements

2.4.2 System Architecture

2.5 Development Methodology

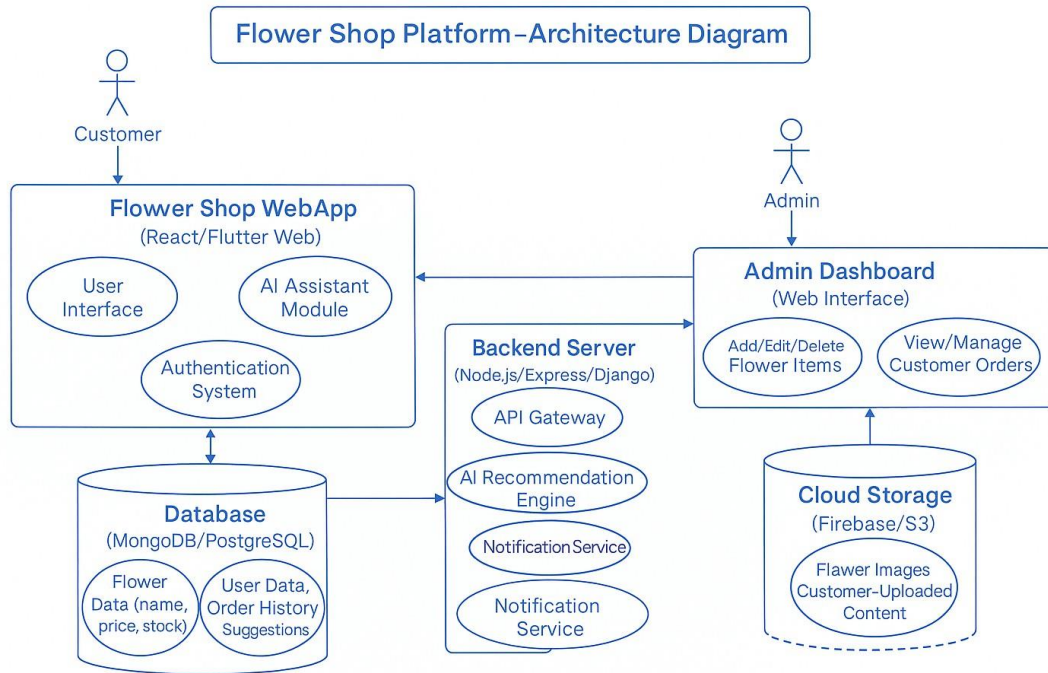


Figure 3-1: Architecture Diagram

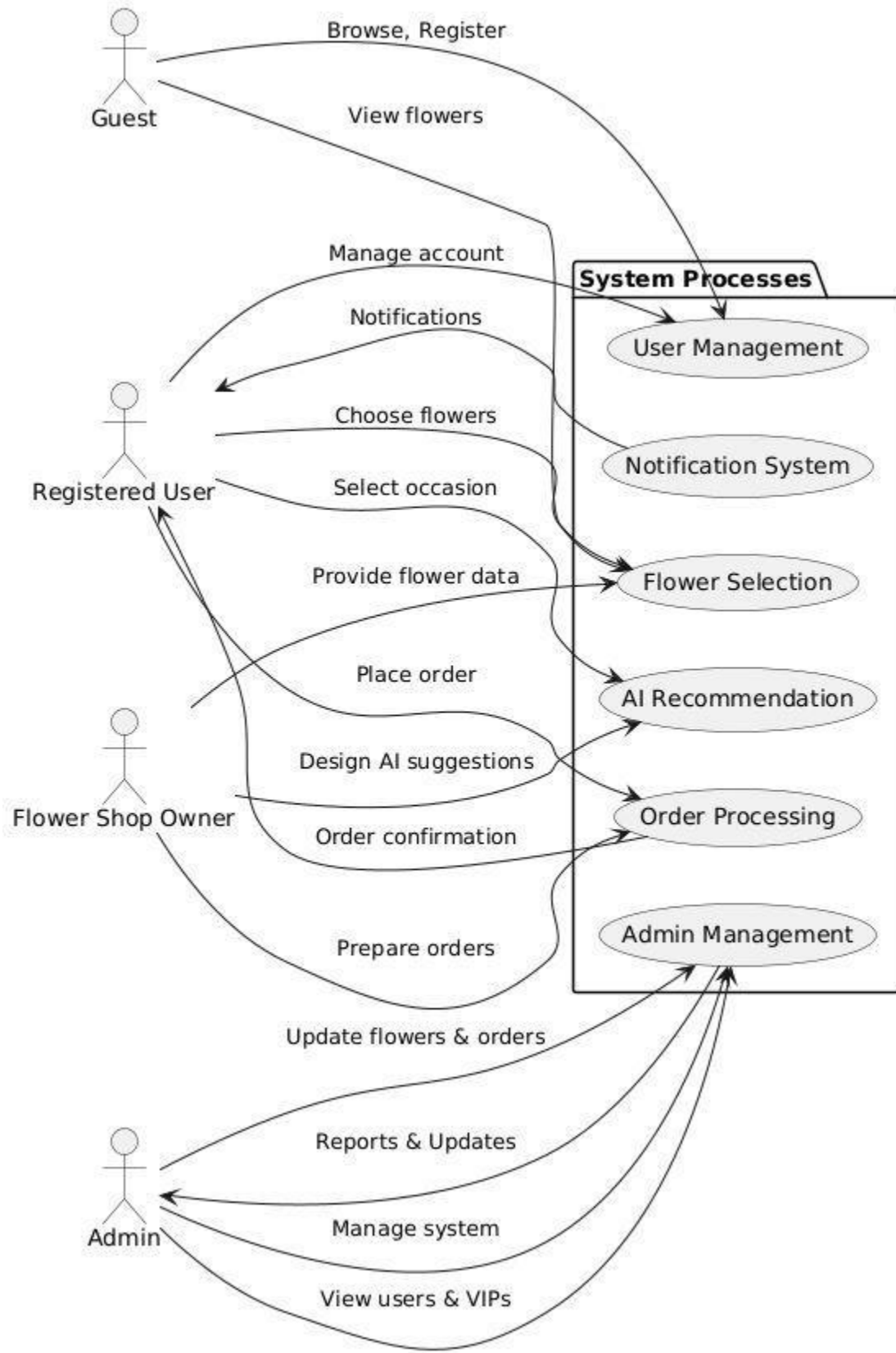
2.5.1 DFD

Data Flow Diagram (DFD) – Level 0

The DFD Level 0 shows the data flow in the main processes of the system.

Processes:

- User Management
- Flower Selection
- AI Recommendation
- Order Processing
- Notification System
- Admin Management



2.5.2 Use Case Diagrams

The Use Case Diagram shows the interaction of the different user types with the system.

Actors: Visitor, Customer, Administrator

Main Use Cases:

Browse Flowers

Customize Bouquet

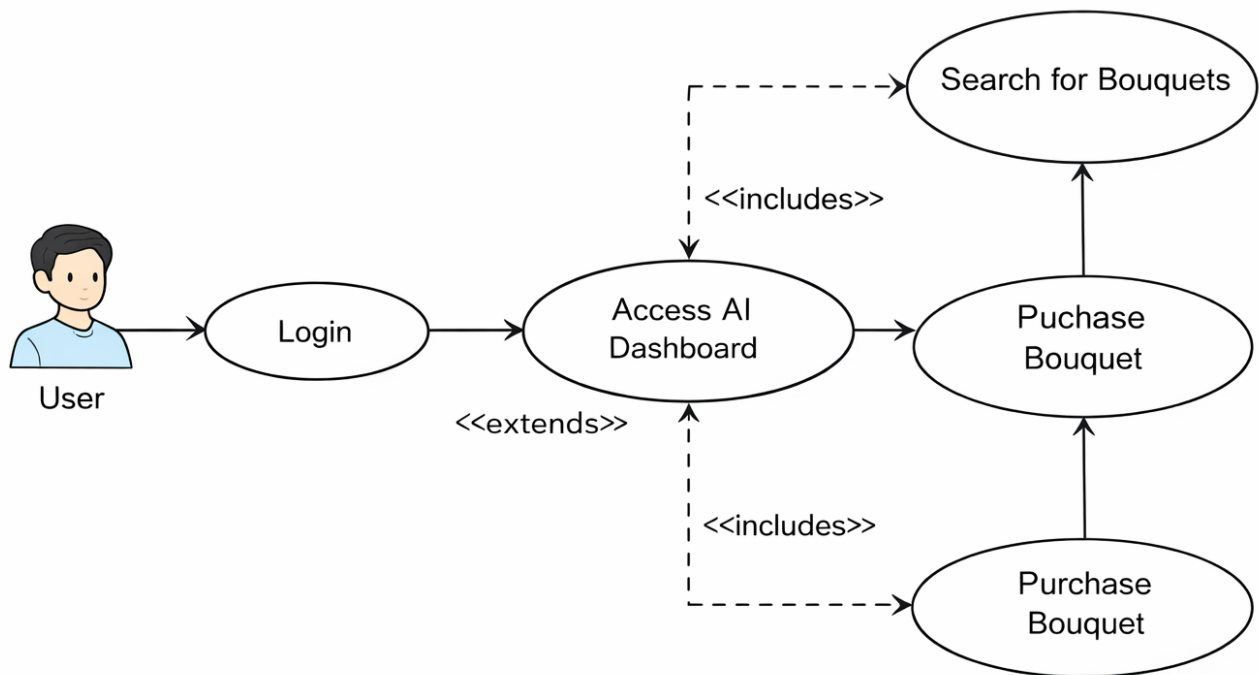
Use AI Assistant

Place Order

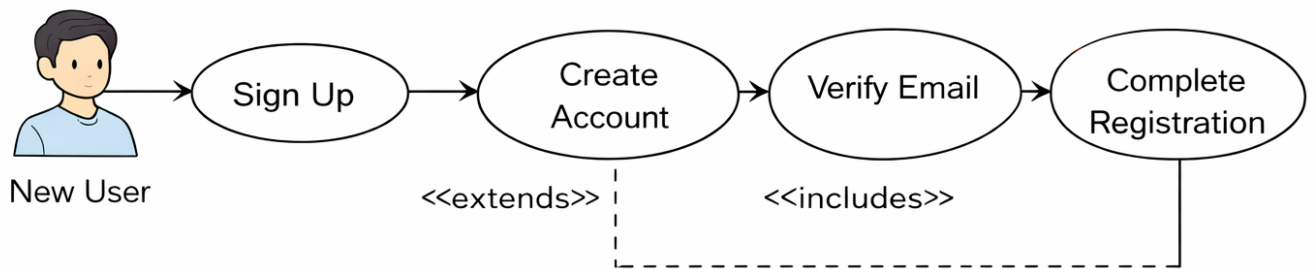
Receive Notifications

Manage Inventory (Admin only)

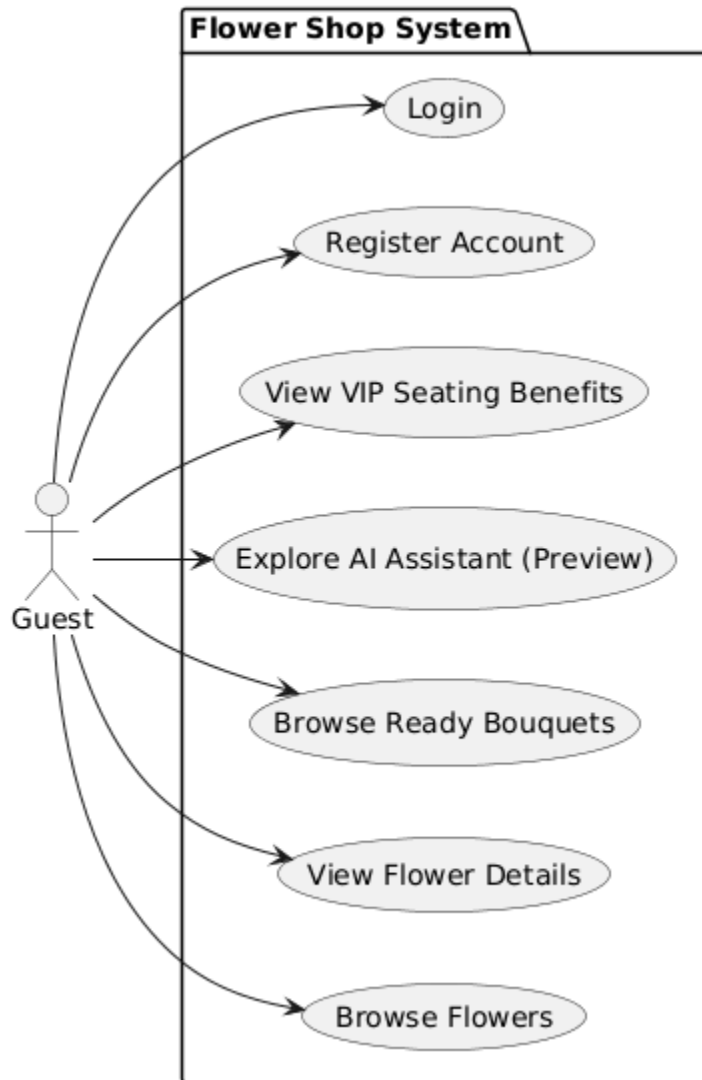
Manage Orders (Admin only)



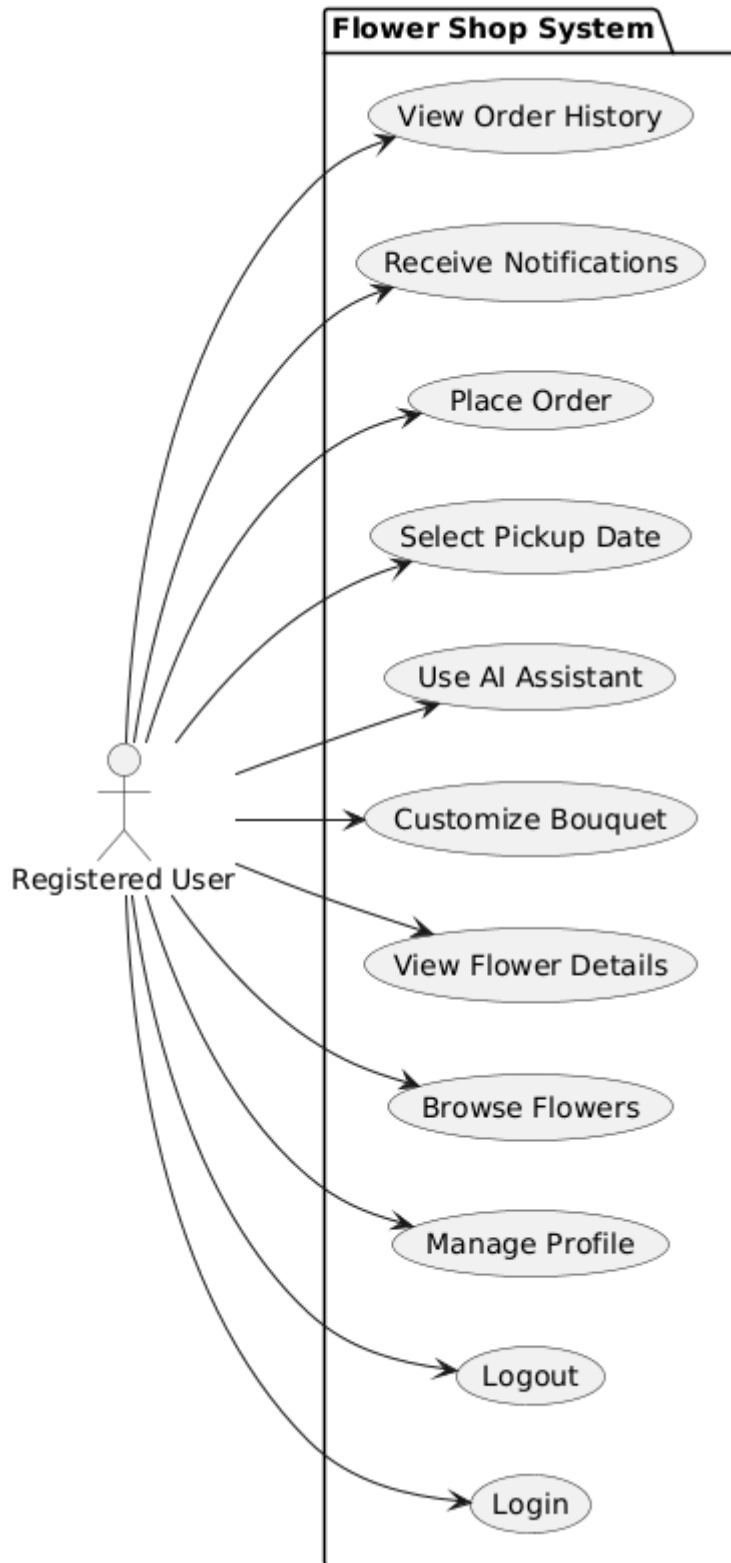
Use Case Diagram



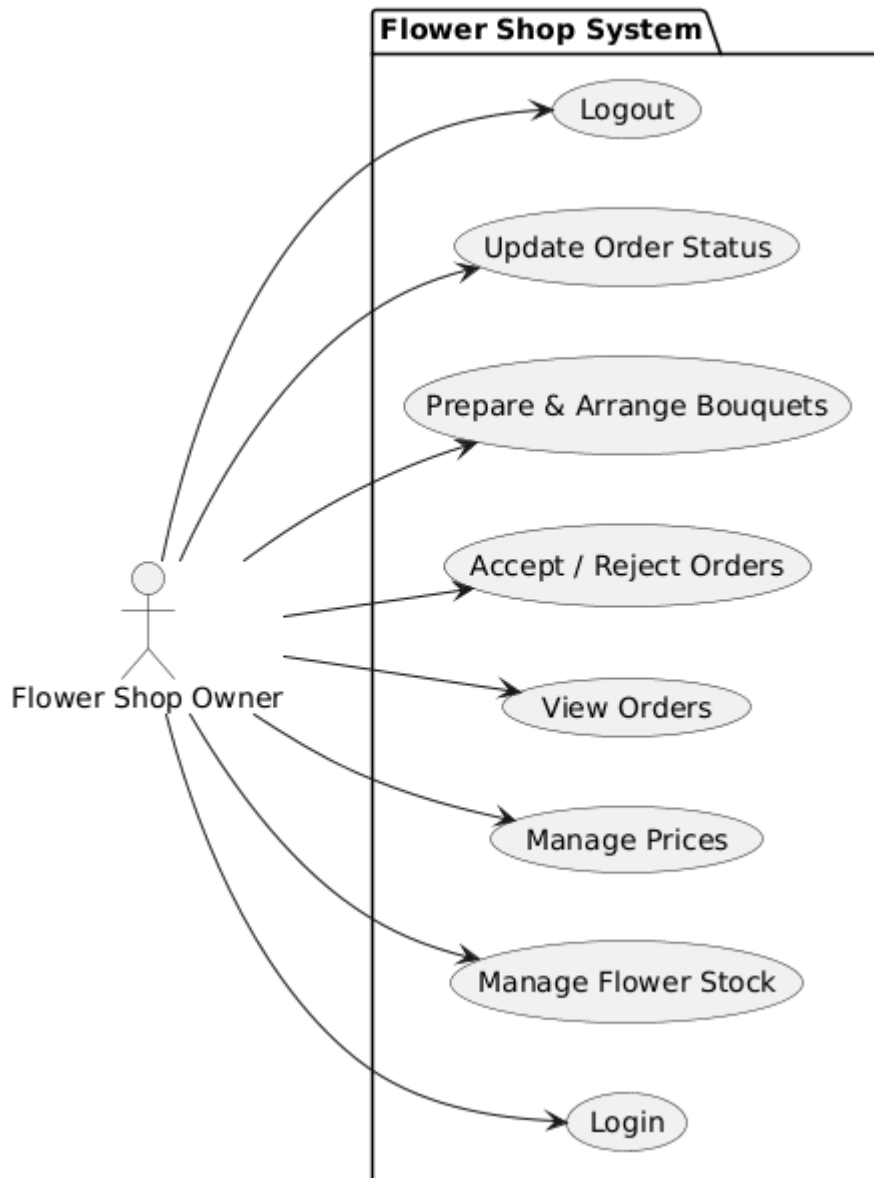
Use Case Diagram



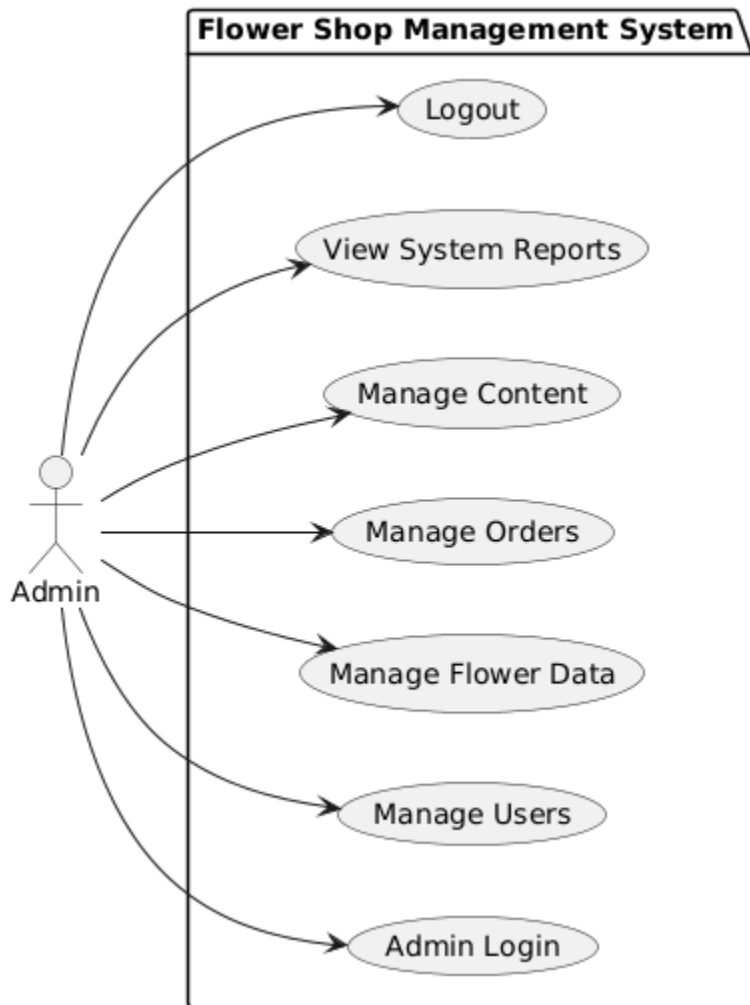
Guest is the name given to the user who is using the platform of the flower shop without login access. This user, referred to as the Guest, is able to browse through flowers on offer, view their prices, check pre-designed bouquets, and understand the benefits offered by VIP membership. Nonetheless, the Guest does not get to design or order bouquets and place orders until an account is made and accessed. This is the initial interface with the platform and is intended to be easy and compelling to access.



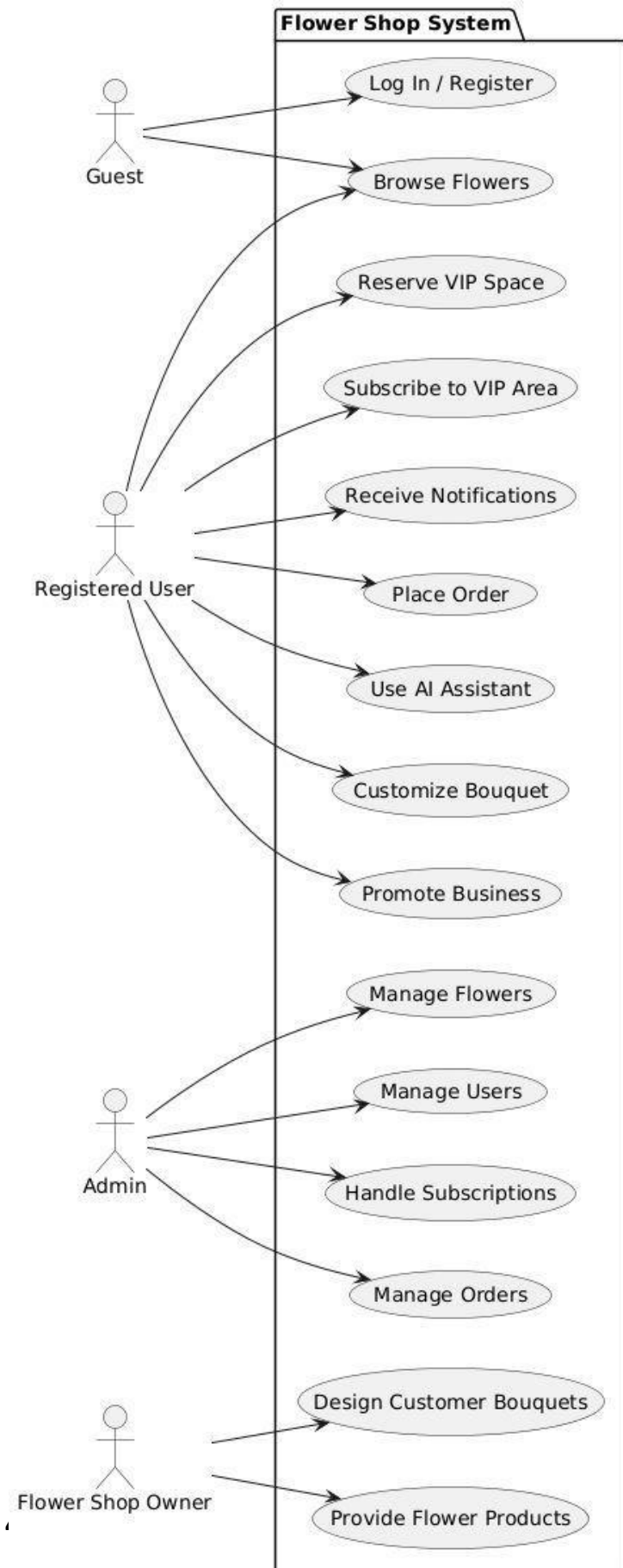
Registered User: This refers to a customer who has made an account and accessed the flower shop. The registered user has the ability to browse the flowers available, access detailed information on flowers, and also create bouquets of different flowers of their choice. The user will then pay for the flowers based on the bouquet calculated automatically. The user will then make an order of the bouquet and the date they will collect the flowers. The registered user will, therefore, have the ability to manage their personal details and the ability to access the ordering history. The admin and flower shop owner will manage the general and flower shop management.



The Flower Shop Owner will be tasked with managing the day-to-day activities of the flower shop through the system. The tasks include logging into the system, managing flower inventory and prices, viewing and confirming customer order requests, and then organizing flower arrangements based on order requests or AI suggestions, and then completing order statuses until they are picked up. This position would be tasked with order fulfillment and ensuring the smooth operation of the flower shop and would have no system admin management functionality.



The Admin takes care of flower shop platform management and maintenance from the backend perspective. Once he/she has logged into his/her admin dashboard, he/she performs operations of user accounts, flower data and contents of the system, and handling customers' orders by altering their statuses. The Admin takes care of the performance of the system as well. Admin users don't have operations of ordering and altering flower bouquets. This is because their sole responsibility is managing and controlling systems.

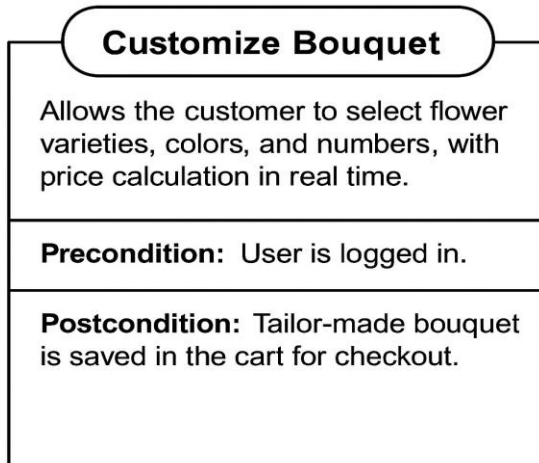


2.5.3 Use Case Description (Detailed Use Cases)

Allows the customer to select flower varieties, colors, and numbers, with price calculation in real time.

Precondition: User is logged in.

Postcondition: Tailor-made bouquet is saved in the cart for checkout.



2.5.4 Sequence Diagram

Scenario: Placing an Order

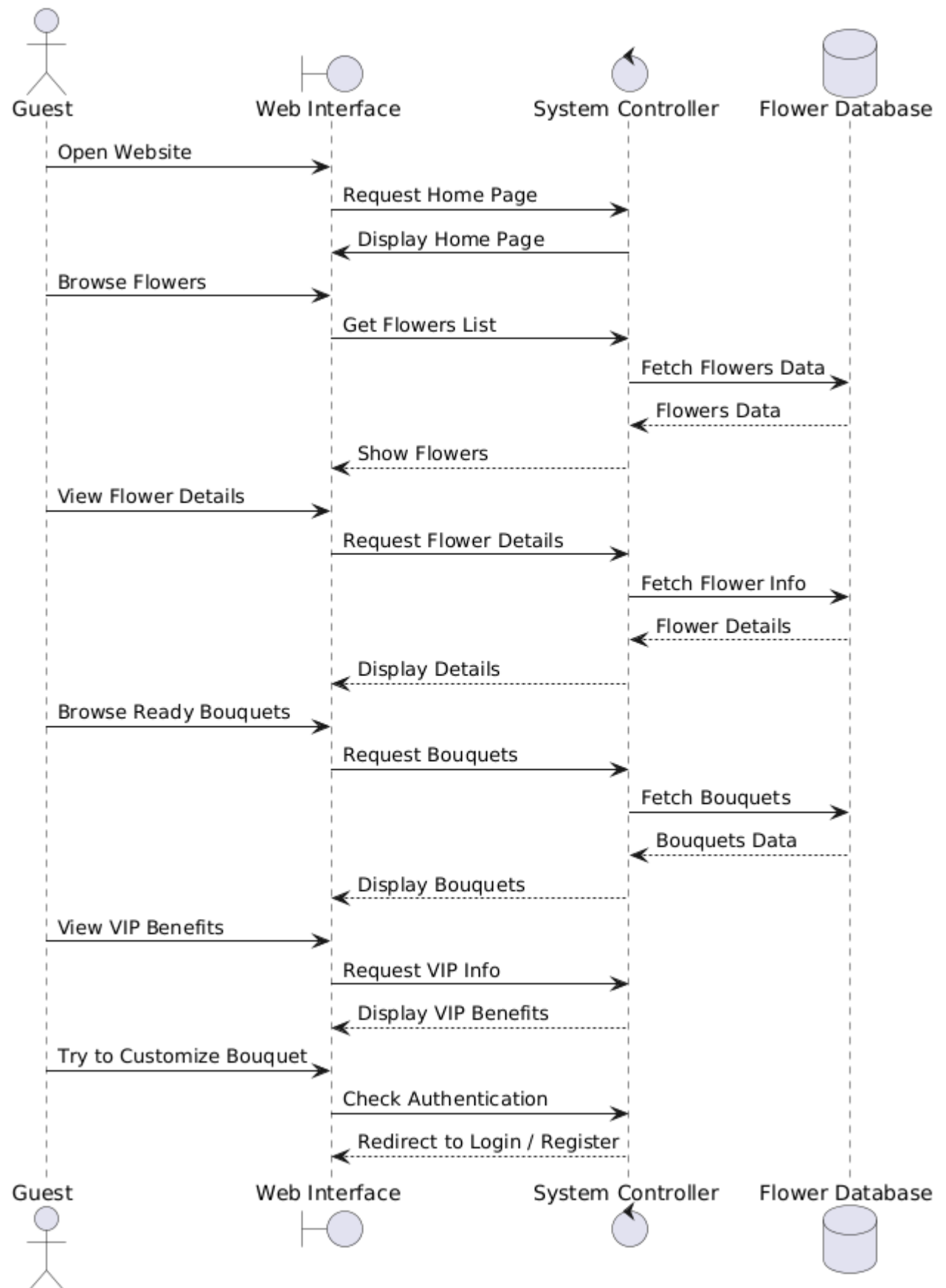
Customer logs in.

Customer select occasion → AI Assistant recommends.

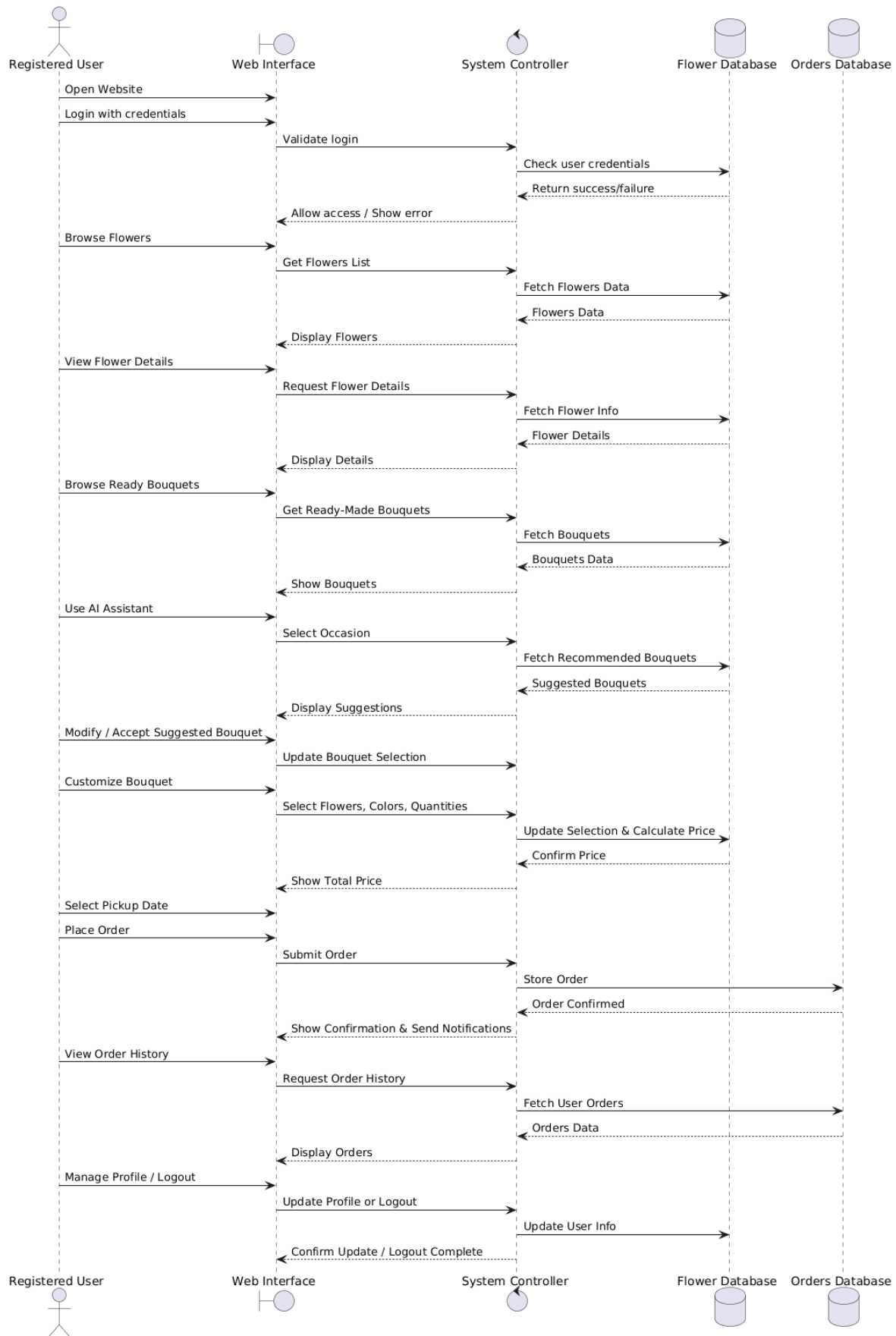
Customer completes order and selects pickup date.

System confirms with a notification.

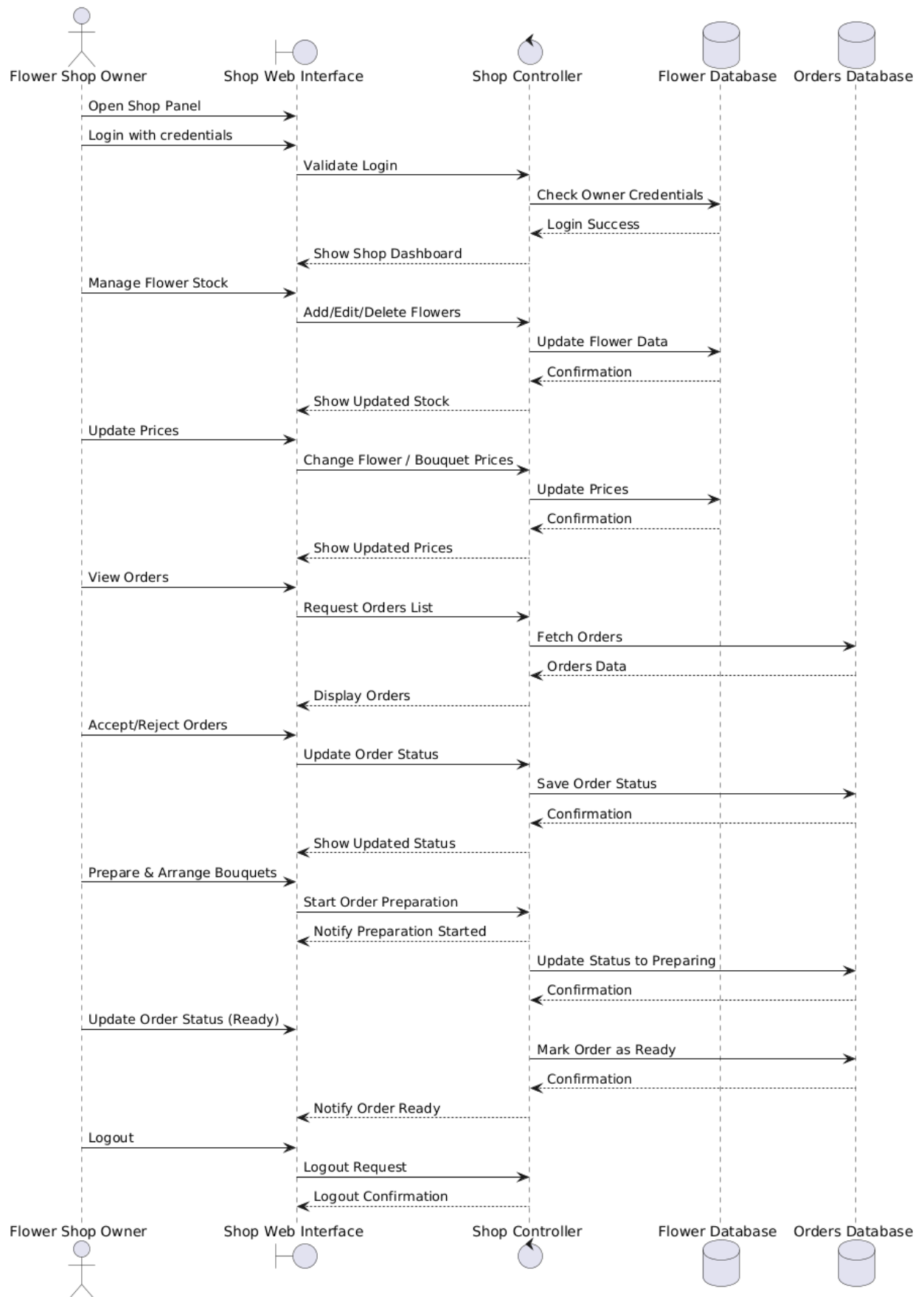
Admin alerted with order details.



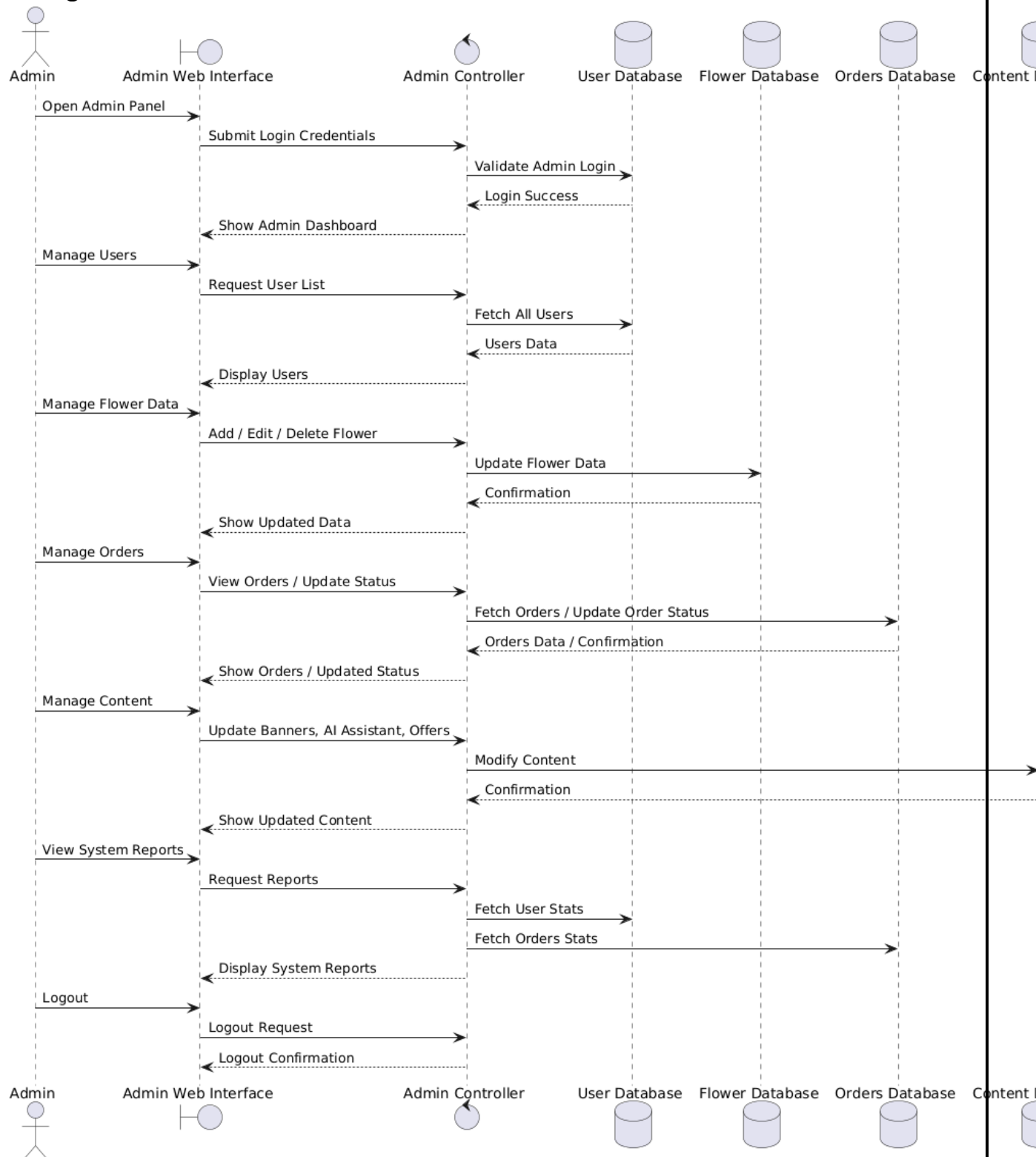
The Guest sequence diagram describes the interaction of a visitor with the system without logging in. The guest can view flowers, view details, view ready-made bouquets, and view the benefits of VIP. However, when a guest attempts to customize a bouquet or purchase one, the system redirects him to register and log in.



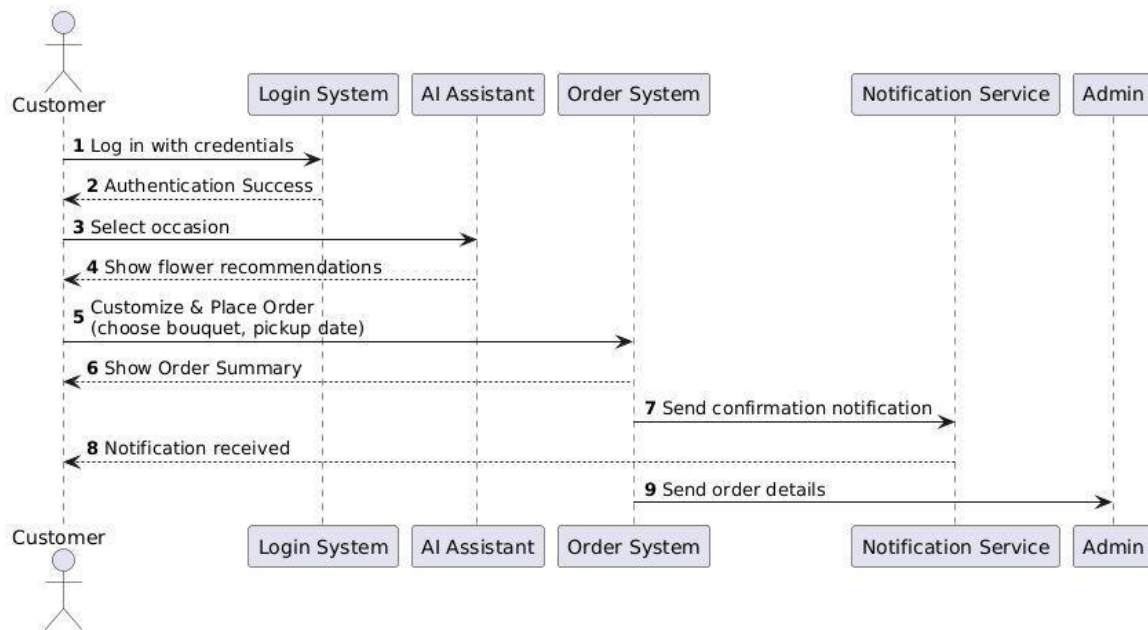
Registered User sequence diagram presents the interaction of a particular customer who has logged in to the system: browsing flowers, showing details, ready-made bouquet exploration, and AI Assistant suggestion, customization of the bouquet, choosing the quantity and pickup date, ordering, order history, and profile management. The system's interaction involves fetching data, price calculations, order storage, and notifications throughout the process.

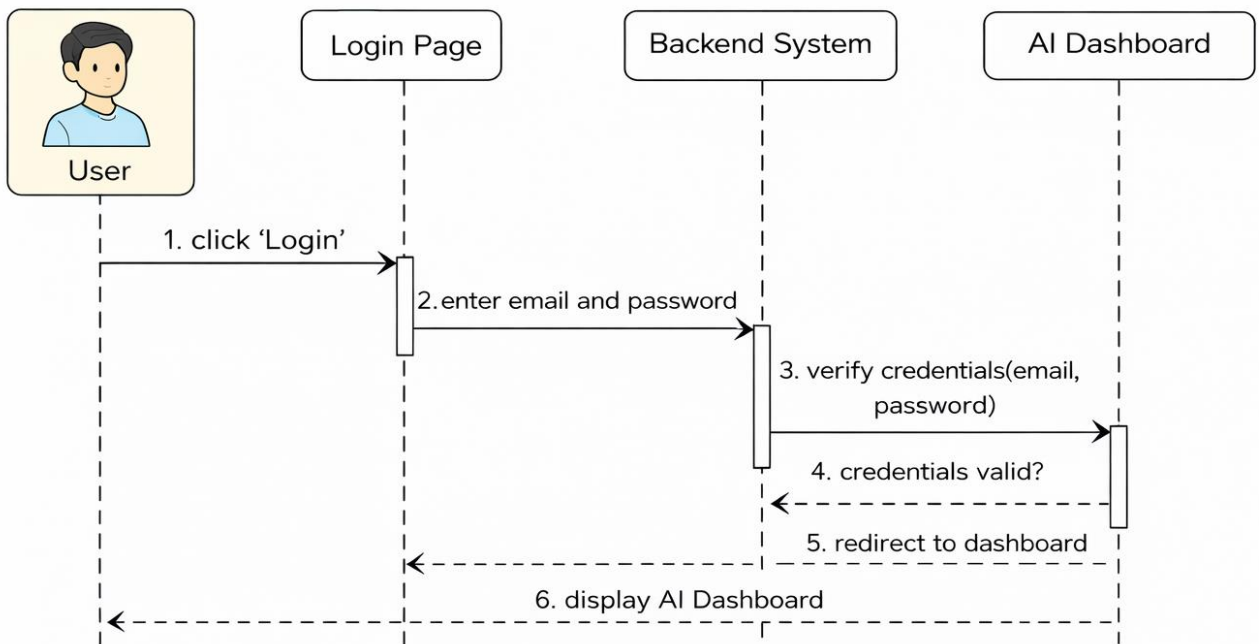


The sequence diagram for the Flower Shop Owner illustrates the involvement of the owner in the management of the shop. The owner logs in, maintains the flow of flowers and their prices, views and confirms or declines customer orders, prepares and arranges bouquets, maintains order statuses, and logs out.

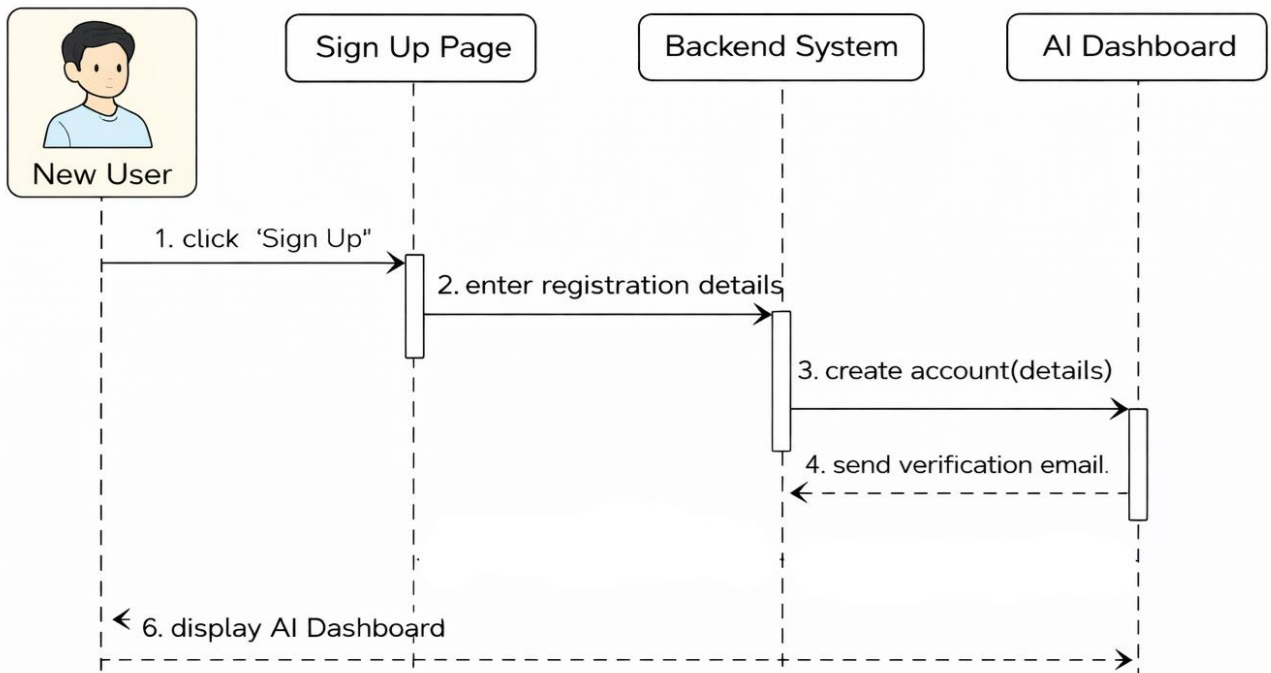


The sequence diagram, 'Admin', shows how the admin interacts with the platform. The admin logs in, goes to the dashboard, administers users, updates flower information, administers orders, administers the system content, examines reports, and successfully logs out. An admin is responsible for ensuring the platform is running smoothly and is organized without any involvement in orders, being a customer.





Login Sequence Diagram



Sign Up Sequence Diagram

2.6 Summary

The system analysis paves the way for creating an intelligent, user-friendly flower shop system beyond the limitations of traditional shopping. Through structured modeling and clear user role definitions, we render the system secure, effective, and tailored to customer satisfaction and shop management needs.

Chapter Four: System Design

The design of the project explains how each section of the problem was formulated. Research and investigate each question or hypothesis and if appropriate, it identify all variables and how they are manipulated.

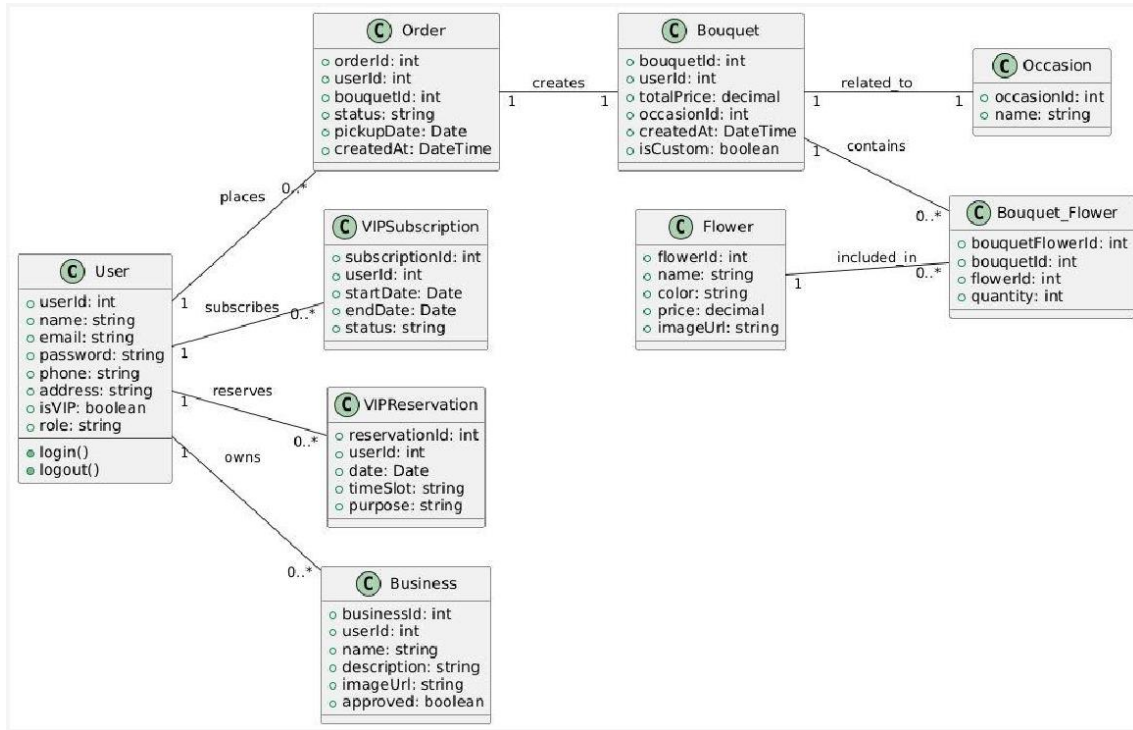
4.1 Introduction

In the design phase of the proposed system, we focused on designing the structural model that meets the specifications outlined in the system requirements.

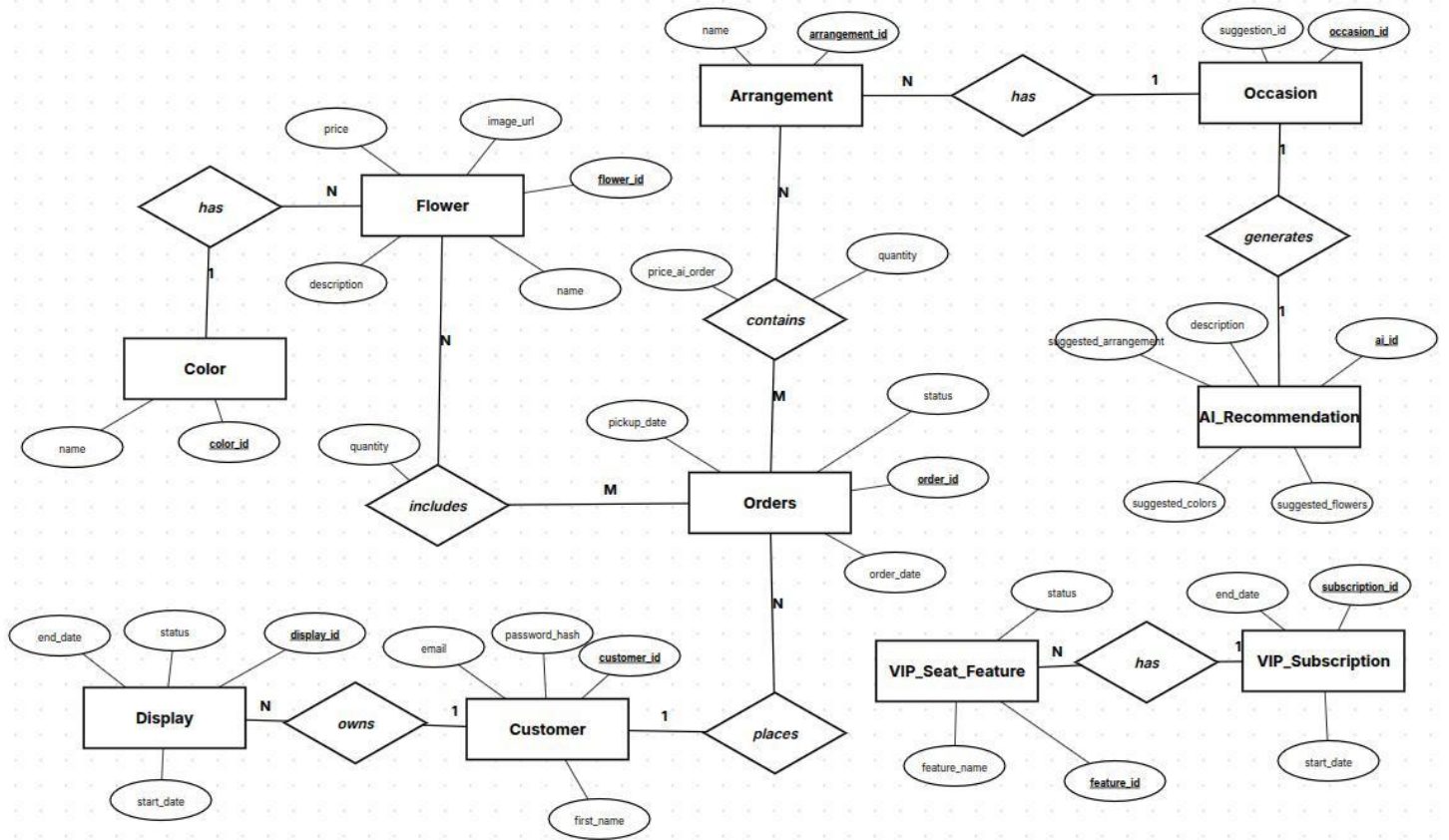
This chapter includes two main sections. In the first section, we will clarify the Class Diagram that represents the static structure of the system by displaying the classes present in the project, their properties, functions, and the relationships between them. Class diagrams are widely used in software development to visualize and design the structure of the software system.

The second section will address the database model by creating the Entity-Relationship Diagram (ER Diagram). This diagram was designed based on the review and analysis of Data Flow Diagrams (DFDs) and the class diagram to identify all relevant entities in the system and their relationships. It also illustrates the primary and foreign keys that connect these entities.

4.2 Class Diagram



4.3 ER diagram



Flower Table:

Attribute	Type	Constraint
flower_id	int	Primary Key
name	varchar(100)	Not Null, Unique
description	text	
price	decimal(10,2)	Not Null
image_url	varchar(255)	

Color Table:

Attribute	Type	Constraint
color_id	int	Primary Key
name	varchar(50)	

Arrangement Table:

Attribute	Type	Constraint
arrangement_id	int	Primary Key
occasion_id	int	Foreign Key → Occasion(occasion_id)
name	varchar(50)	

Occasion Table:

Attribute	Type	Constraint
occasion_id	int	Primary Key
suggestion_id	int	

AI Recommendation Table:

Attribute	Type	Constraint
ai_id	int	Primary Key
occasion_id	int	Foreign Key → Occasion(occasion_id)
suggested_flowers	TEXT	
suggested_colors	TEXT	
suggested_arrangement	TEXT	
description	TEXT	

Customer Table:

Attribute	Type	Constraint
customer_id	int	Primary Key
first_name	varchar(50)	
email	varchar(50)	
password_hash	varchar(500)	

Order Table:

Attribute	Type	Constraint
<u>order_id</u>	int	Primary Key
<u>customer_id</u>	int	Not Null, FK → Customer(<u>customer_id</u>)
<u>order_date</u>	datetime	Not Null
<u>pickup_date</u>	date	Not Null
status	<u>varchar(20)</u>	Not Null
<u>suggestion_template_id</u>	int	FK → <u>Bouquet_Template(template_id)</u> , nullable

Order_Item Table:

Attribute	Type	Constraint
order_id	int	Foreign Key
arrangement_id	int	Foreign Key
quantity	int	
price_ai_order	DECIMAL	

Order_Item_Flower Table:

Attribute	Type	Constraint
order_id	int	Foreign Key
flower_id	int	Foreign Key
quantity	INT	

VIP_Subscription Table:

Attribute	Type	Constraint
subscription_id	int	Primary Key
start_date	date	
end_date	date	

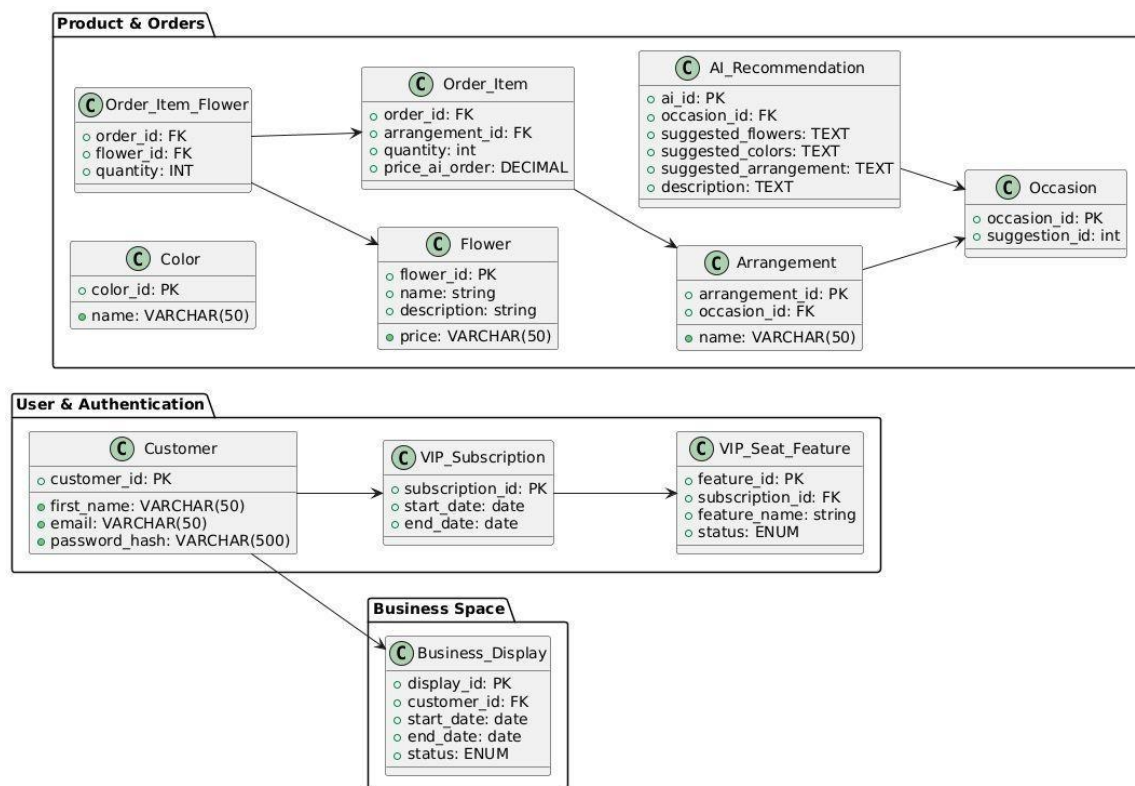
VIP_Seat_Feature Table:

Attribute	Type	Constraint
feature_id	int	Primary Key
subscription_id	int	Foreign Key
feature_name	string	
status	ENUM	

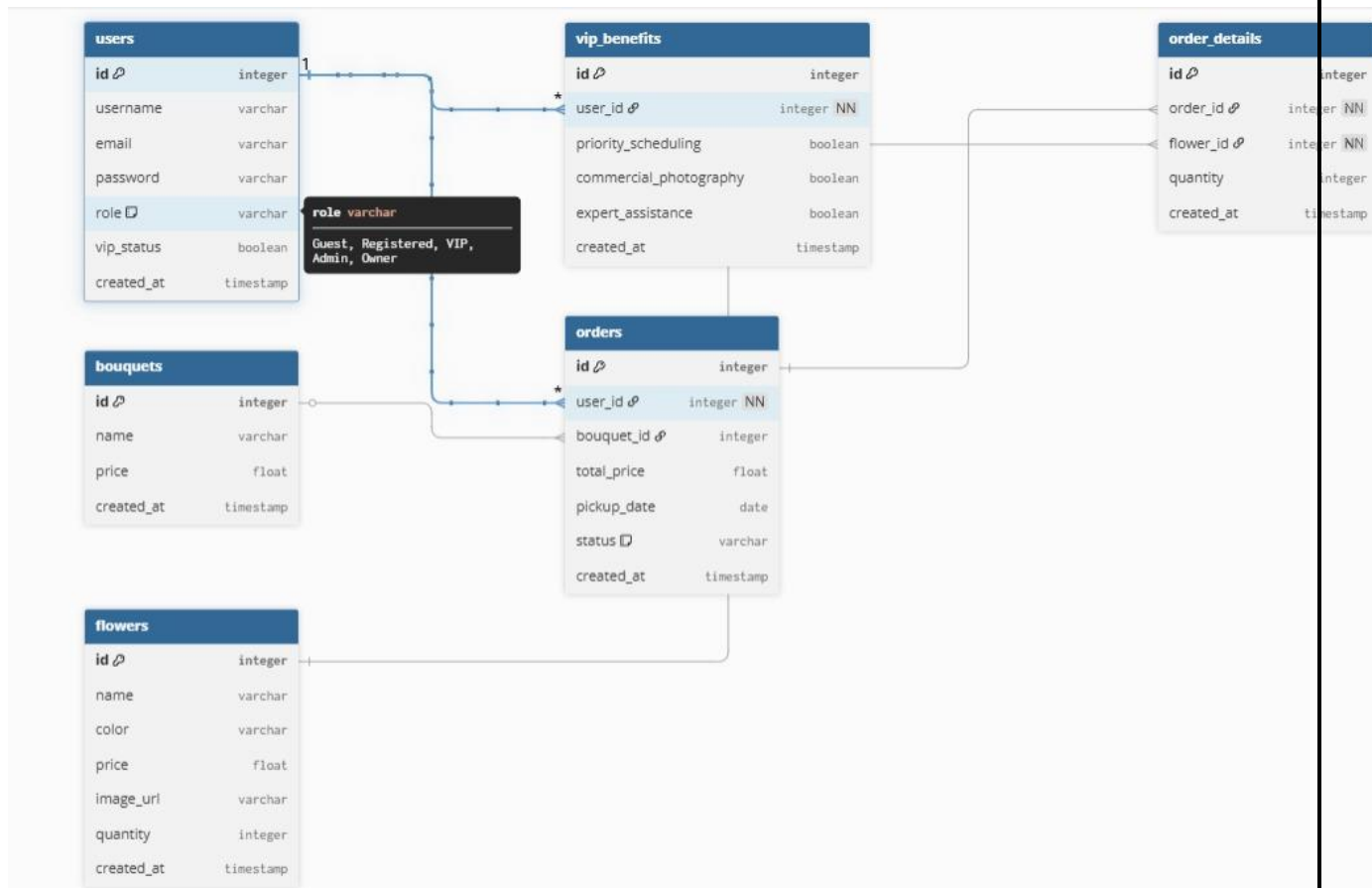
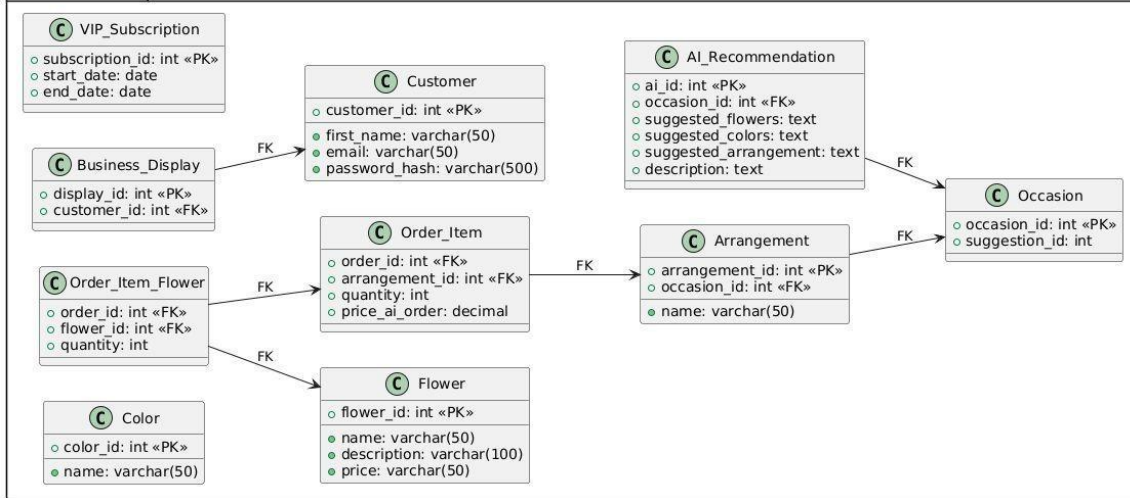
Business_Display Table:

Attribute	Type	Constraint
display_id	int	Primary Key
customer_id	int	Foreign Key
start_date	date	
end_date	date	
status	ENUM	

4.4 Mapping Data Base

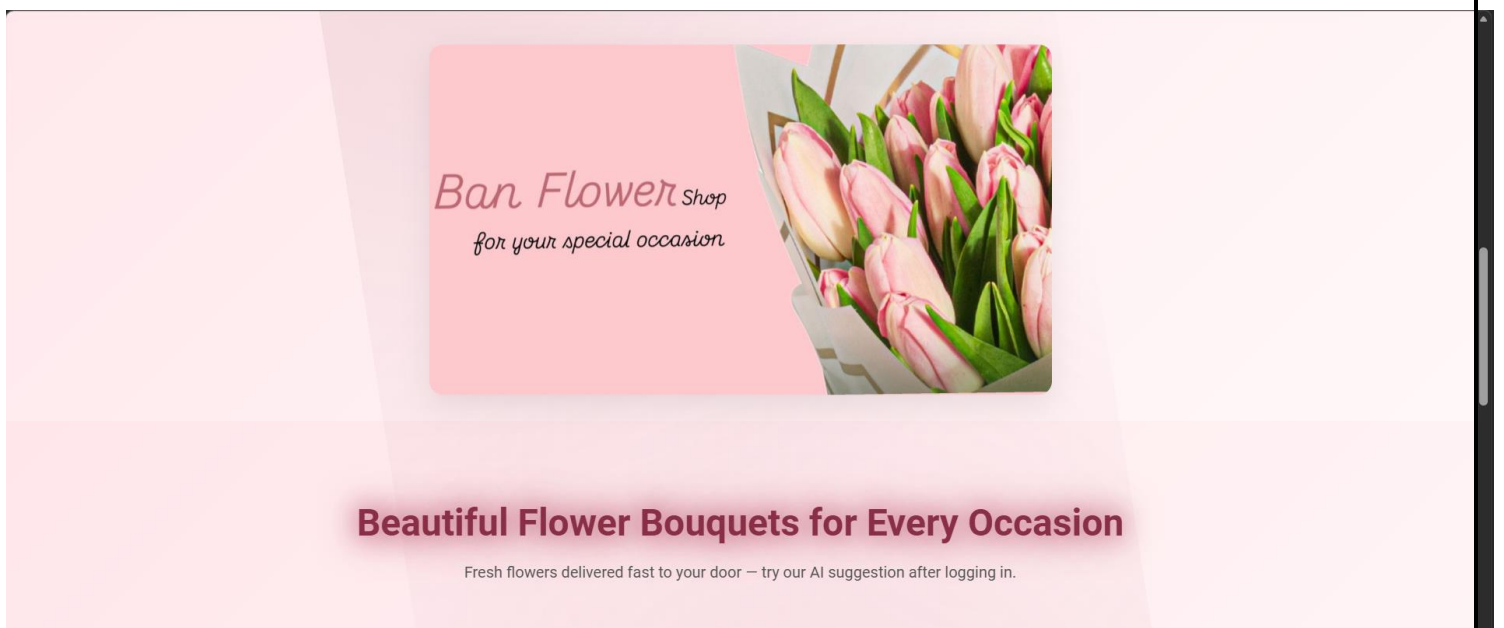
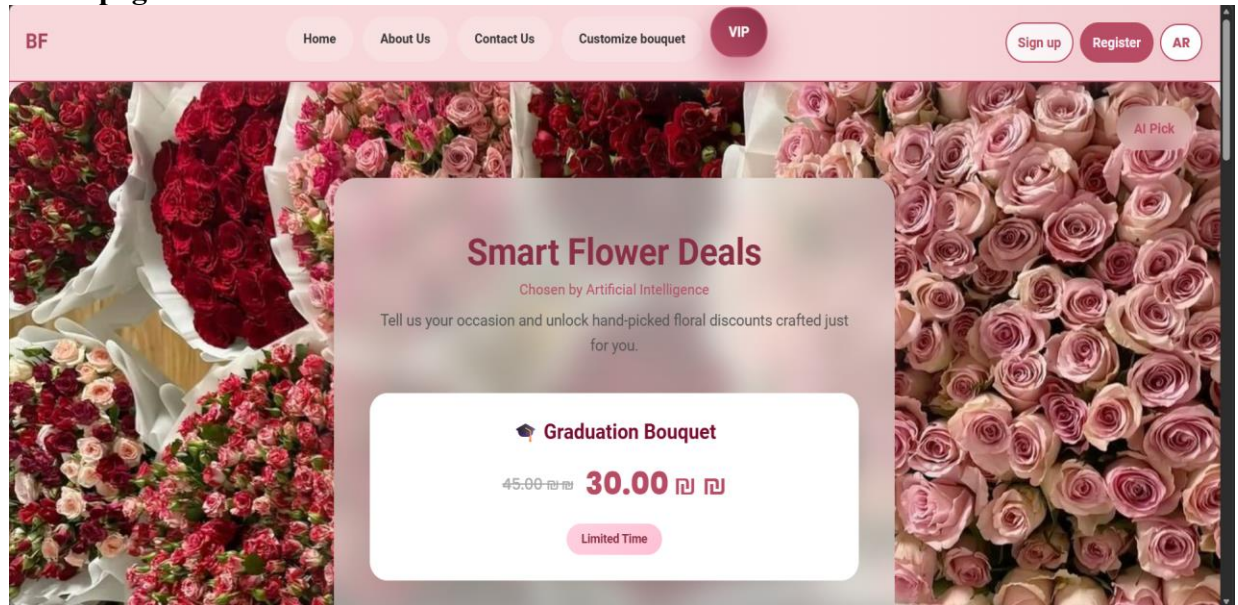


Flower Shop DB



4.5 Interface Design

4.5.1 Home page



Beautiful Flower Bouquets for Every Occasion

Fresh flowers delivered fast to your door — try our AI suggestion after logging in.



Balloons



Baskets



Bouquets



Chocolate Bouquets



Gifts



Graduation



Money Bouquets

127.0.0.1:5501/flowers project - raheeq/flowers project - Copy (3) (1)/flowers project - Copy (3)/flowers project - Copy/flowers project - Copy/balloons1.html

VIP Seating Area

Enjoy our cozy and aesthetic VIP seating area, available by subscription. Perfect for studying, relaxing, or working in a floral paradise.

VIP subscribers can also showcase their small businesses with product photography support in a beautiful floral setting.

[Subscribe Now](#)



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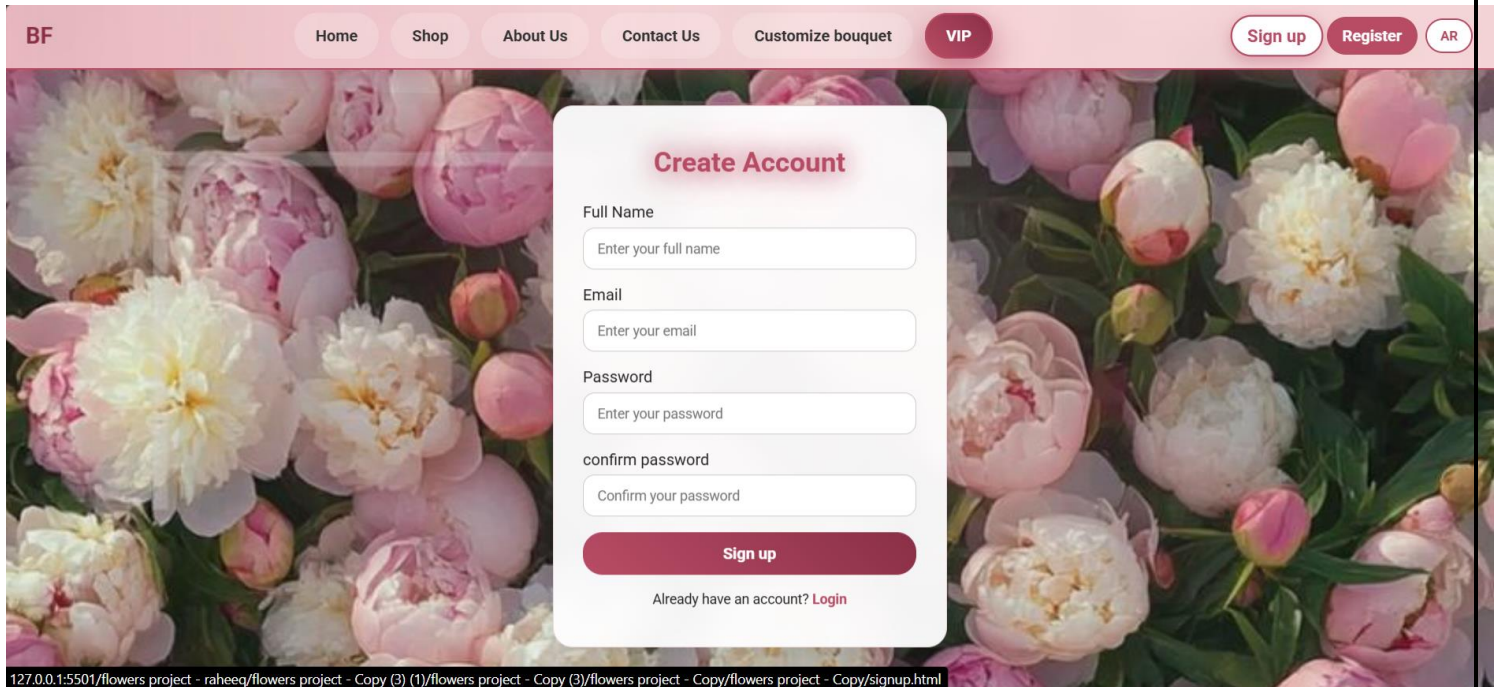
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Colors Symbolic

Flower Care

4.5.2 Sign in page



The sign up page features a pink header with the logo 'BF' and navigation links: Home, Shop, About Us, Contact Us, Customize bouquet, and a highlighted VIP button. On the right, there are buttons for Sign up, Register, and AR. The main content area has a background of pink and white peonies. A central white form titled 'Create Account' contains fields for Full Name, Email, Password, and confirm password, followed by a Sign up button and a link to Login for existing users.

BF Home Shop About Us Contact Us Customize bouquet VIP Sign up Register AR

Create Account

Full Name
Enter your full name

Email
Enter your email

Password
Enter your password

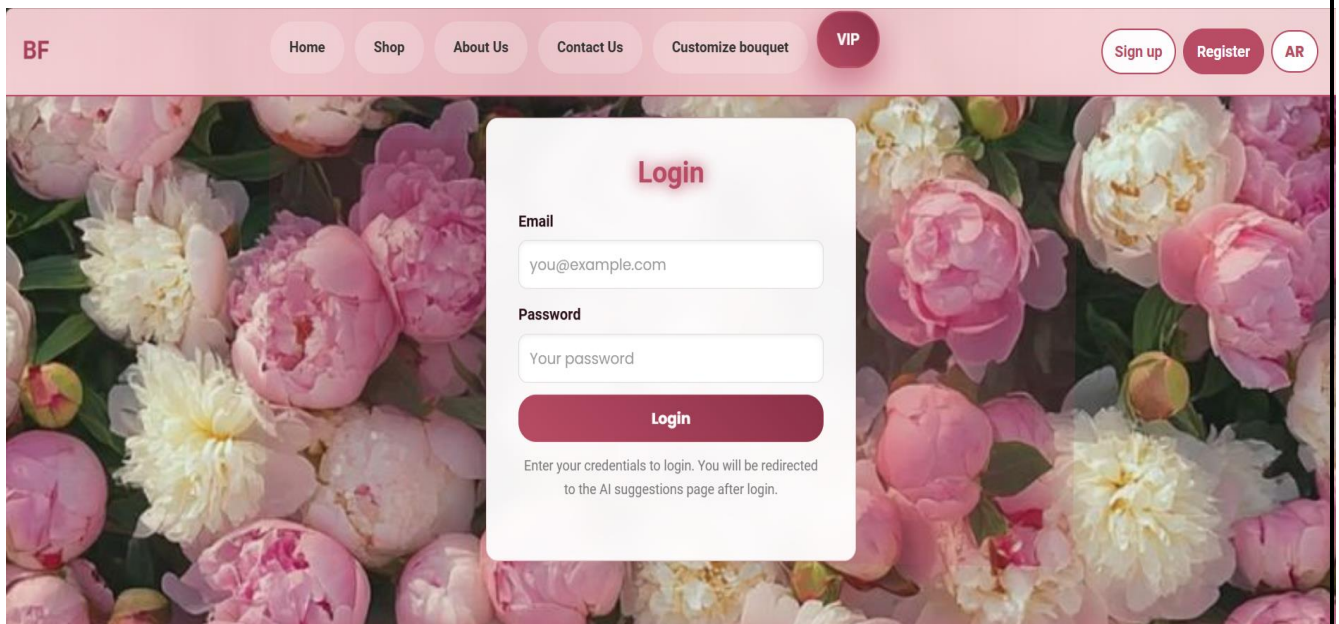
confirm password
Confirm your password

Sign up

Already have an account? [Login](#)

127.0.0.1:5501/flowers project - raheeq/flowers project - Copy (3) (1)/flowers project - Copy (3)/flowers project - Copy/flowers project - Copy/signup.html

4.5.3 Login page



The login page has the same header as the sign up page. The main content area features a background of pink and white peonies. A central white form titled 'Login' contains fields for Email and Password, followed by a Login button. Below the button, a message states: 'Enter your credentials to login. You will be redirected to the AI suggestions page after login.'

BF Home Shop About Us Contact Us Customize bouquet VIP Sign up Register AR

Login

Email
you@example.com

Password
Your password

Login

Enter your credentials to login. You will be redirected to the AI suggestions page after login.

4.5.4 Ready-made Bouquets

Fresh flowers delivered fast to your door — try our AI suggestion after logging in.



Balloons



Baskets



Bouquets



Chocolate Bouquets



Gifts



Graduation



Money Bouquets

BF

Home

About Us

Contact Us

Customize bouquet

VIP

عربي

Cart

Our Products



Classic Bouquet

Price: 120.00 ₪

1

Add to Cart



Romantic Bouquet

Price: 150.00 ₪

1

Add to Cart



Romantic Bouquet

Price: 150.00 ₪

1

Add to Cart



Romantic Bouquet

Price: 150.00 ₪

1

Add to Cart



Classic Bouquet

Price: 120.00 ₪

1

Add to Cart



Romantic Bouquet

Price: 150.00 ₪

1

Add to Cart



Your Cart

Your cart is empty.

Subtotal	0.00 ₪
Tax (5%)	0.00 ₪
Total	0.00 ₪

Clear cart

Checkout



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Our Products



Classic Bouquet

Price: 120.00 ₪

1

Add to Cart



Romantic Bouquet

Price: 150.00 ₪

1

Add to Cart



Romantic Bouquet

Price: 130.00 ₪

1

Add to Cart



Romantic Bouquet

Price: 120.00 ₪

1

Add to Cart



Classic Bouquet

Price: 150.00 ₪

1

Add to Cart



Romantic Bouquet

Price: 130.00 ₪

1

Add to Cart



1

Add to Cart



Romantic Bouquet
Price: 120.00 ₪

1

Add to Cart



Romantic Bouquet
Price: 150.00 ₪

1

Add to Cart

Added ✓

Your Cart

Your cart is empty.

BF

HomeAbout UsContact UsCustomize bouquetVIP

عربيCart

Our Products



Classic Bouquet
Price: \$120.00

1

Add to Cart



Romantic Bouquet
Price: \$150.00

1

Add to Cart



Romantic Bouquet
Price: \$150.00

1

Add to Cart

67



Classic Bouquet

Price: \$120.00

1

Add to Cart



Romantic Bouquet

Price: \$150.00

1

Add to Cart



Romantic Bouquet

Price: \$150.00

1

Add to Cart



Romantic Bouquet

Price: \$150.00

1

Add to Cart



Classic Bouquet

Price: \$120.00

1

Add to Cart



Romantic Bouquet

Price: \$150.00

1

Add to Cart



Romantic Bouquet

Price: \$150.00

1

Add to Cart

Added ✓

Your Cart

Your cart is empty.

Subtotal 0.00 ₺

Tax (5%) 0.00 ₺

Our Products



Classic Bouquet

Price: 120.00 ₪

1

Add to Cart



Romantic Bouquet

Price: 150.00 ₪

1

Add to Cart



Romantic Bouquet

Price: 150.00 ₪

1

Add to Cart



Romantic Bouquet

Price: 150.00 ₪

1

Add to Cart



Classic Bouquet

Price: 120.00 ₪

1

Add to Cart



Romantic Bouquet

Price: 150.00 ₪

1

Add to Cart



Added ✓
Your Cart

Your cart is empty.

Subtotal	0.00 ₪
Tax (5%)	0.00 ₪
Total	0.00 ₪

Clear cart

Checkout



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Cart

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Classic Bouquet

Price: 120.00 ₪

1

Add to Cart



Romantic Bouquet

Price: 150.00 ₪

1

Add to Cart



Romantic Bouquet

Price: 150.00 ₪

1

Add to Cart



Romantic Bouquet

Price: 150.00 ₺

1

Add to Cart



Classic Bouquet

Price: 120.00 ₺

1

Add to Cart



Romantic Bouquet

Price: 150.00 ₺

1

Add to Cart



Romantic Bouquet

Price: 150.00 ₺

1

Add to Cart



Romantic Bouquet

Price: 150.00 ₺

1

Add to Cart

Added ✓

Your Cart

Your cart is empty.

Subtotal

0.00 ₺

Our Products



Classic Bouquet

Price: 120.00 ₪

1

Add to Cart



Romantic Bouquet

Price: 150.00 ₪

1

Add to Cart

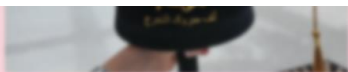


Romantic Bouquet

Price: 150.00 ₪

1

Add to Cart



Romantic Bouquet

Price: 150.00 ₪

1

Add to Cart



Romantic Bouquet

Price: 150.00 ₪

1

Add to Cart



Classic Bouquet

Price: 120.00 ₪

1

Add to Cart



Romantic Bouquet

Price: 150.00 ₪

1

Add to Cart

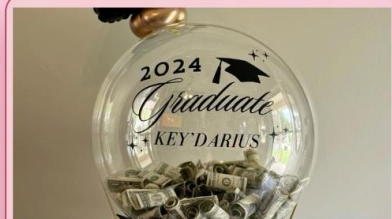


Romantic Bouquet

Price: 150.00 ₪

1

Add to Cart



Romantic Bouquet

Price: 150.00 ₪

1

Add to Cart



Romantic Bouquet

Price: 150.00 ₪

1

Add to Cart



Romantic Bouquet

Price: 150.00 ₪

1

Add to Cart

Added ✓

Your Cart

Your cart is empty.

Subtotal 0.00 ₪

Tax (5%) 0.00 ₪

BF

Home

Shop

About Us

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Classic Bouquet

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Add to Cart



Romantic Bouquet

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Romantic Bouquet

Price: 150.00 ₪

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Classic Bouquet

Price: 120.00 ₪

1

Add to Cart



Romantic Bouquet

Price: 150.00 ₪

1

Add to Cart



Romantic Bouquet

Price: 150.00 ₪

1

Add to Cart



Added to cart

Your Cart

Your cart is empty.

Subtotal 0.00 ₪

Tax (5%) 0.00 ₪

Total 0.00 ₪

Clear cart

Checkout



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MEGA FLOWER DEALS

Limited time offers – Don't miss the bloom 🌸

Shop Offers Now

Today's Hot Deals



Graduation Bouquet

~~\$40~~

\$30

Buy Now



Romantic Roses

~~\$35~~

\$25

Buy Now



Birthday Basket

~~\$50~~

\$40

Buy Now

\$40

\$30

Buy Now

\$35

\$25

Buy Now

\$50

\$40

Buy Now





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Register

AR

Let's make something beautiful together

Have a question, custom order request, or want to book a VIP table at our studio? Send us a message — we usually reply within a few hours.

call: +970-595-789760

view location

Tip: for faster replies, include your preferred delivery date and the occasion.

Contact Us

Fill out the form below and we will get back to you soon.

Name

Email

Phone

Preferred delivery date (optional)

mm/dd/yyyy



Number of bouquets(optional)

Message

127.0.0.1:5501/flowers project - raheeq/flowers project - Copy (3) (1)/flowers project - Copy (3)/flowers project - Copy/flowers project - Copy/contact.html

76

Message

Send message

Reset

We'll never share your information. By sending you agree to our terms.

BF

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Customize Your Bouquet

24 / 7 delivery • Build and preview your bouquet live

Refined • Aesthetic • Fast

Search species or color (e.g. rose, pink)

Totals update automatically

All

Pink

Red

White

Yellow

Purple

Tip: choose a species, then select a color inside the card

Available Species

Each species can come in multiple colors – pick the color you want before adding.

Rose

Classic, timeless species with long stems.

20.00 ₪ ₪

Available colors:

1

Add to bouquet

127.0.0.1:5501/flowers project - raheeq/flowers project - Copy (3) (1)/flowers project - Copy (3)/flowers project - Copy/flowers project - Copy/customize_bouquets1.html

77



Tulip

Bright and cheerful, great for celebrations.

Available colors:

-

1

+

12.00 ₪ ₪

Add to bouquet



Lily

Elegant and fragrant — ideal for formal arrangements.

Available colors:

-

1

+

18.00 ₪ ₪

Add to bouquet



Sunflower

Sunny and bold — makes any bouquet pop.

Available colors:

-

1

+

10.00 ₪ ₪

Add to bouquet



Orchid

Exotic, premium flower for luxury bouquets.

35.00 ₪ ₪

Your Bouquet

Live preview & totals

Subtotal

0.00 ₪

Wrapping

0.00 ₪

Tax (5%)

0.00 ₪

Total

0.00 ₪

Your name

Phone or email

Notes for the bouquet (colors, message...)

Order Now

Save Preset (local demo)

24 / 7 Service

We deliver round-the-clock for all orders.

Need help? +970-595789760

[Order Now](#)

[Save Preset \(local demo\)](#)

24 / 7 Service
We deliver round-the-clock for all orders.

Need help? +970-595789760



Contact US

+970-595789760
Working hours:
Saturday to Friday 24 hours

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Design
Prototyping
Development features
Design systems
Collaboration features
Design process
FigJam

Help

Delivery Info
Frequently Asked Questions-FAQ
Colors Symbolic
Flower Care

BF

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Welcome to Our Exclusive VIP Lounge

Subscribe now to enjoy a luxurious, cozy space for studying, relaxing, or promoting your small business in a stylish floral setting.

[Subscribe Now](#)

BF

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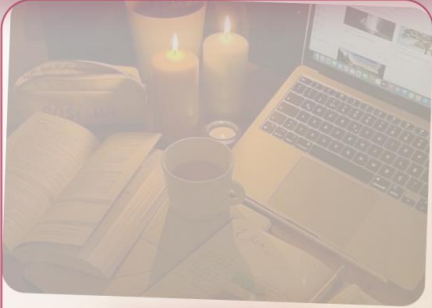
Customize bouquet

VIP

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AR



Cozy Study Spots

Quiet and elegant spaces surrounded by flowers to boost focus and creativity.



Promote Your Business

Showcase your products in a luxurious floral environment for professional photos.



Product Photography

Capture stunning visuals of your products with a classy floral backdrop.

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Relaxation Space

Unwind in a sophisticated atmosphere with comfy seating and aesthetic flowers.

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Relaxation Space

Unwind in a sophisticated atmosphere with comfy seating and aesthetic flowers.

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About Us

We are a passionate Computer Science team, dedicated to learning, innovating, and creating impactful projects while exploring the world of technology.

Our Team

127.0.0.1:5501/flowers project - raheeq/flowers project - Copy (3) (1)/flowers project - Copy (3)/flowers project - Copy/flowers project - Copy/aboutus.html

Our Team



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Yaqeen Al-Saadi

Email: yaqeenalsadi@gmail.com
Phone: +970 595 789 760



Sahera Daraghme

Email: saheradaraghme@gmail.com
Phone: +972 59-286-6417

Our Values

Innovation

We strive to create new ideas and

Collaboration

Working together as a team to

Continuous Learning

We keep improving our skills and

Our Values

Innovation

We strive to create new ideas and
solutions in technology projects.

Collaboration

Working together as a team to
achieve excellence.

Continuous Learning

We keep improving our skills and
exploring new technologies.

Achievements



Completed advanced
courses in Web
Development, AI, and
python.



Won second place in
university IT club and
coding competitions.



Published multiple projects
on GitHub and collaborated
with peers.



Received recognition for
innovative and creative
class projects.

Why Choose Us

Strong Technical Skills

We are Computer Science students with solid foundations in programming, problem solving, and modern technologies.

Continuous Learning

We constantly improve our skills through courses, workshops, and hands-on projects in different tech fields.

Teamwork & Collaboration

We believe in collaboration, sharing ideas, and working together to achieve high-quality results.

Award-Winning Projects

Our projects have been recognized in university competitions and hackathons for creativity and innovation.

Creative & Innovative Mindset

We focus on building creative solutions that combine good design with efficient functionality.

Commitment & Reliability

We are committed to deadlines, quality work, and delivering reliable solutions every time.

Contact Us

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saheradaraghmeh@gmail.com

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[LinkedIn Profile](#)



Contact US

+970-595789760

Working hours:

Saturday to Friday 24 hours

Explore

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Development features

Design systems

Collaboration features

Help

Delivery Info

Frequently Asked Questions-FAQ

Colors Symbolic

Flower Care

Smart Bouquet Suggestion (AI)

Occasion

Other occasion

Tone / Style

Delivery date (optional)

Quantity

Color preferences ☐ ☐ ☐ ☐ ☐

Suggestions are local-only demo list.

Smart Bouquet Suggestion (AI)

Occasion

Other occasion

Tone / Style

Delivery date (optional)

Quantity

Color preferences ☐ ☐ ☐ ☐ ☐

Suggestions are local-only demo list.

4.5.5 Delivery information page

Let's make something beautiful together

Have a question, custom order request, or want to book a VIP table at our studio? Send us a message — we usually reply within a few hours.

call: +970-595-789760

[view location](#)

Tip: for faster replies, include your preferred delivery date and the occasion.

Contact Us

Fill out the form below and we will get back to you soon.

Name

Email

Phone

Preferred delivery date (optional)

mm/dd/yyyy



Number of bouquets(optional)

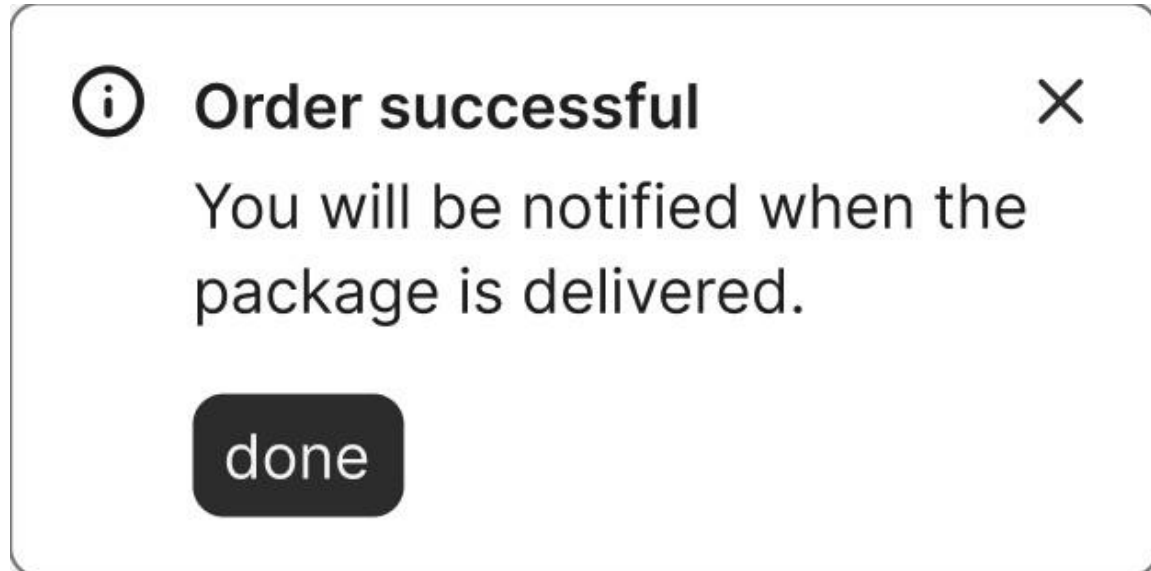
Message

[Send message](#)

[Reset](#)

We'll never share your information. By sending you agree to our

4.5.6 Notification



4.5 Story Design

Story Overview:

Our project is an internet interactive application for a flower arrangement store within the city of Jenin that is meant to simplify the process of purchasing and arranging flowers online in an enjoyable way. The site offers users the ability to browse through flower categories, choose colors and quantities, create the bouquet shape, as well as the ability to become a premium VIP seat subscription user for personal use or to showcase small business products in a comfortable and inviting environment. The overall goal is to enhance the user experience and provide innovative solutions that meet flower hobbyists, customers, and small business owners' requirements.

Characters :

Registered User:

A registered user created within the app can personalize bouquets, select flowers and quantities, input personal details, and make orders.

Flower Shop Owner:

Operates the store, controls flower stock and prices, accepts orders from the application, and provides flower arranging services.

Admin:

Operates the application from the backend side, managing the user base, orders, and content to ensure an organized and efficient user run.

VIP Subscriber:

A registered user who has availed of the private seating space within the store, either for studying, working, or showcasing products. They have the additional benefit of priority scheduling, commercial photography, and expert assistance.

Visitor:

Any guest who enters the app without logging in. They are able to navigate flowers, view prices and bouquets, and find out about VIP seating benefits, but cannot buy or create bouquets until they create an account and log in. Guests are the first entry point into the app, so an easy and attractive browsing experience has been designed for them.

Implementation:

The UI was developed using the newest web development and UI design software with focus on a hassle-free and user-friendly experience. Another system was integrated to manage flower and bouquet information, orders, and VIP seat subscribers.

Key Features:

Browse and Personalize Bouquets:

An entire list of types of flowers with the capability to select the quantity and color and put them together in a personalized bouquet.

- **VIP Seating System:**

The ability to subscribe for premium seats within the store, with some usage options such as studying or product display.

- **Order Management and Notifications:**

Tracking of order status in real-time with order confirmation and update notifications.

- **Easy-to-Use Interface:**

A simple, visually appealing and age-friendly design with ease of navigation and purchase for the customer.

Conclusion:

This project is a qualitative innovation in the flower industry in Jenin by offering a combined digital experience with easy purchasing, customization, and innovative additional services such as VIP seats. The project would improve the relationship between the flower stores and the clients and help small business owners by providing an excellent exhibition platform in a professional and appealing environment.

4.6 Summary

The aim of this chapter was to provide an overview of the system design stage of our flower shop web application, the major design factors: class diagram, database design, and user interface design.

In the class diagram, we represented the structural base of the system by introducing core classes such as User, Flower, Bouquet, Order, VIPSubscription, etc., along with their attributes, methods, and how they are connected. It helped to model the object-oriented structure of the system and to develop a clear understanding of how elements cooperate.

Also, we designed the database model by developing the Entity-Relationship Diagram (ERD), which defines all the entities in the system — flowers, users, orders, and VIP functionalities — and defines relationships between them using primary and foreign keys.

Finally, we followed primary UI design principles when designing the user interface in terms of simplicity, usability, accessibility, and visual consistency. The interface facilitates visitors as well as registered members with smooth navigation, and also admins and owners of flower shops with well-organized control panels. Aesthetic and ergonomic experience-based design limitations were also considered to give a pleasant and useful user experience.

References

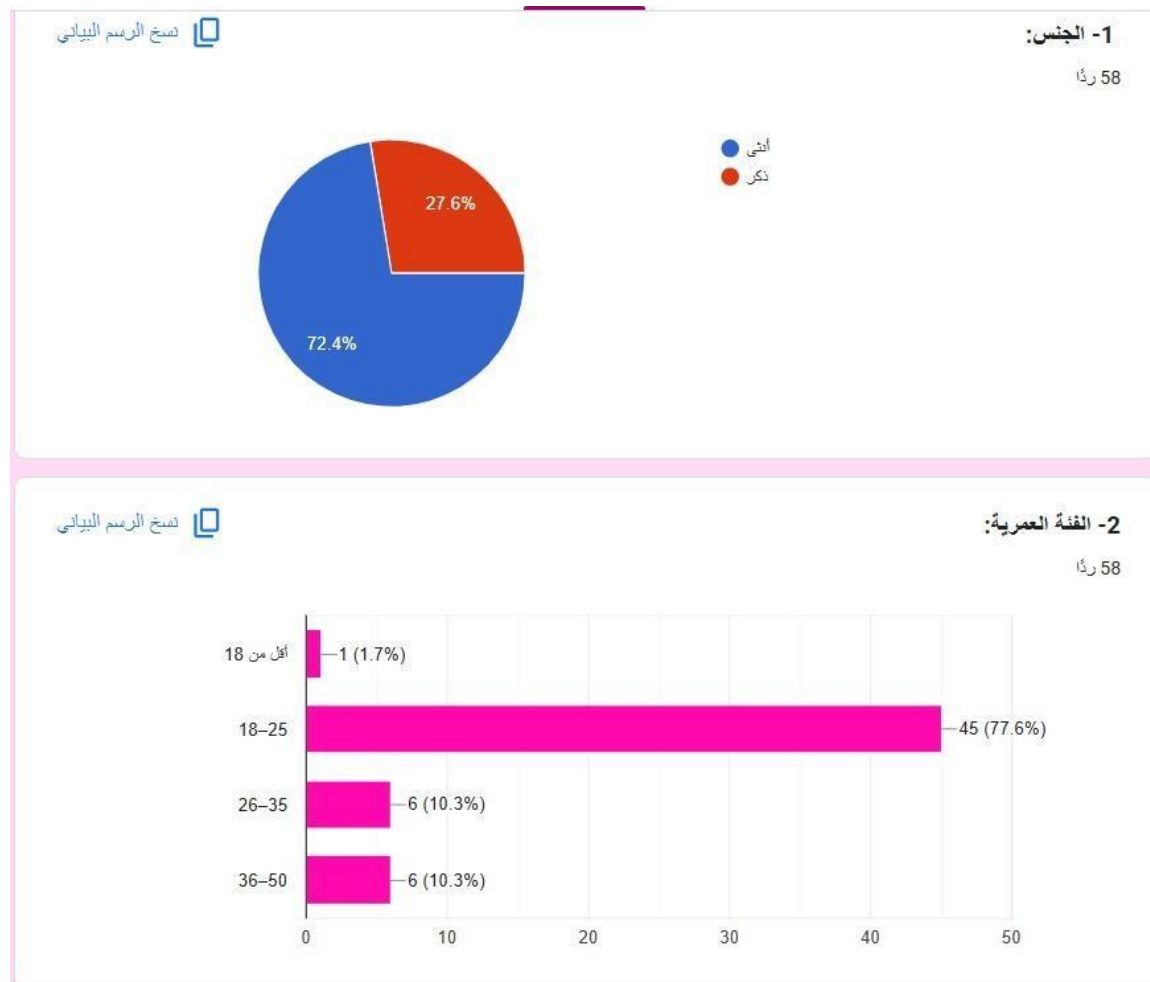
A recommended referencing style is:

Harvard style: <http://www.ucd.ie/t4cms/Guide69.pdf>

1. sys1Floward – <https://www.floward.com/>
2. sys2Ferns N Petals – <https://www.fnp.com/>
3. sys3Interflora – <https://www.interflora.com/>
4. sys41–800–Flowers – <https://www.1800flowers.com/>
5. sys5ProFlowers – <https://www.proflowers.com/>
6. sys6FloraQueen – <https://www.floraqueen.com/>
7. sys7Zhoor Alreef – <https://www.zohooralreef.com/en>
8. sys8Blommarket – <https://www.flowerstosaudiArabia.com/>
9. sys9Talabat – <https://www.talabat.com/>
10. sys10Rose N’ More – <https://www.rosenmore.com/>

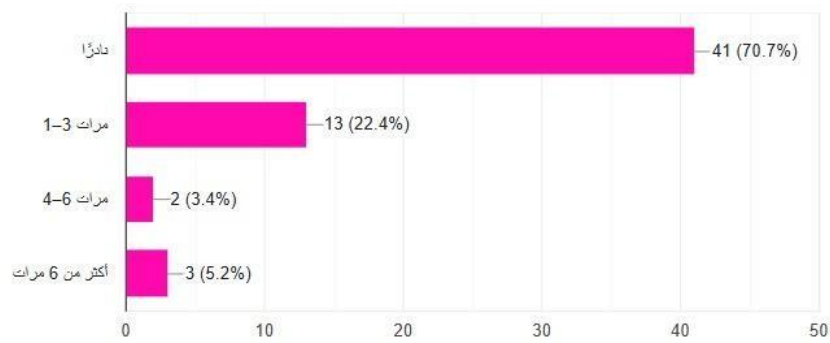
Appendices

Annex 1: Survey for Assessing the Need for the Proposed System.



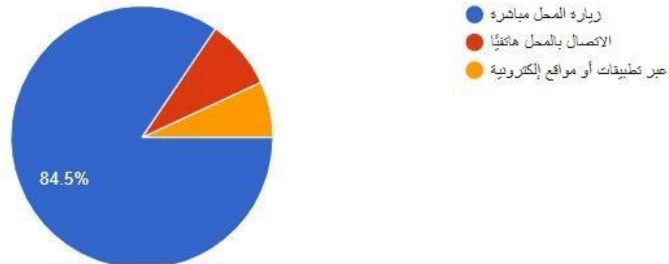
3- كم مرة تقوم بشراء الورود سنويًا؟

ردًا 58



4- ما الطريقة التي تستخدمها غالبًا لشراء الورود؟

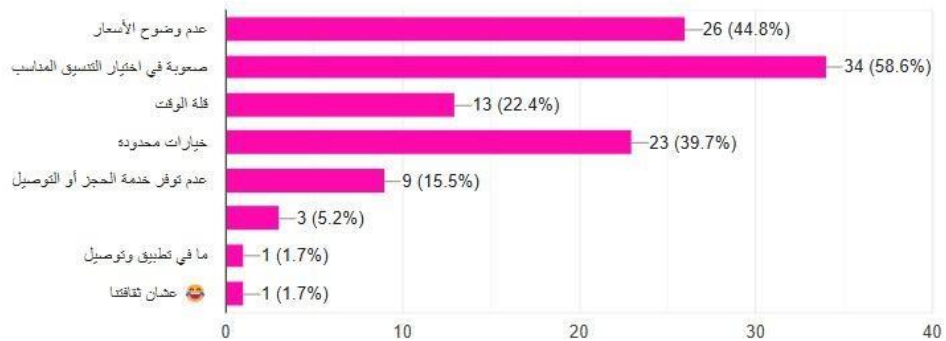
ردًا 58



نسخ الرسم البياني

5- ما هي أبرز الصعوبات التي تواجهها عند شراء الورود؟ (يمكن اختيار أكثر من خيار)

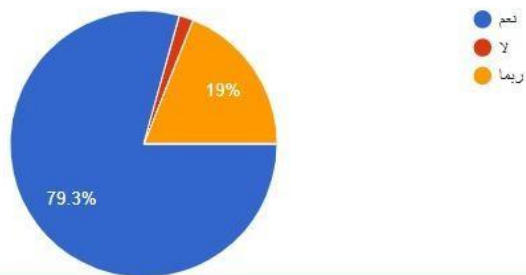
58 ردًا



نسخ الرسم البياني

6- هل ترى أن فكرة تخصيص باقة ورد عبر الإنترنت حسب المناسبة (زفاف، تخرج...) فكرة مفيدة؟

58 ردًا



Drive

نموذج بدون عنوان

الرود 1 الإعدادات

رد واحد

ملخص

الردود

الردود

1- الجنس:

رد واحد

2- اللغة العربية:

رد واحد

100%

الذكور

النساء

نسخ الرسم البياني

9- هل تفضل أن تُتاح لك حرية تعديل الباقة المقترحة أو تصميم باقتك من الصفر؟

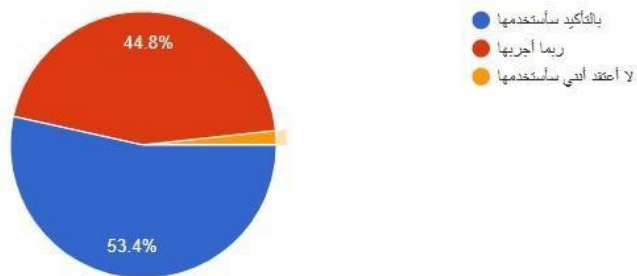
58 ردًا



نسخ الرسم البياني

10- ما مدى استعدادك لاستخدام هذه المنصة عند توفرها؟

58 ردًا



ملاحظات إضافية أو اقتراحات:

12 ردًا

.

لا يوجد

بالتوفيق 🌟

Null

فكرة ممتازة

لا

لا يوجد

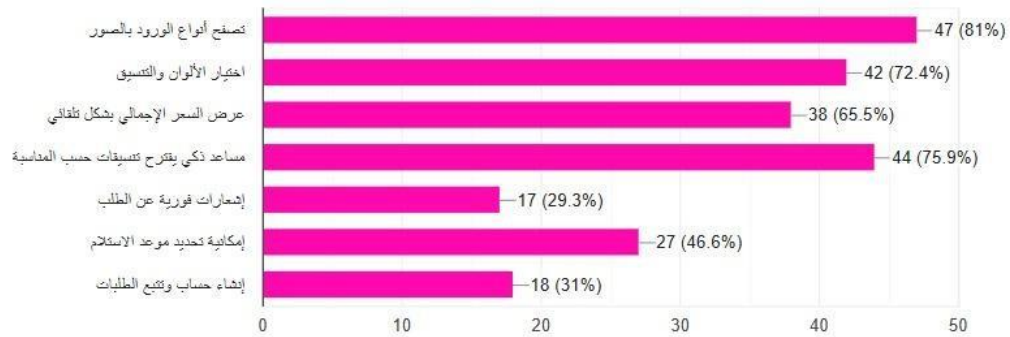
لما تتعرض بأقة الورد تكون بالوالدها الطبيعية بدون فلتر و قريية كثير من الحقيقة لانو اهم اشي الثقة بيهيك شغلات

يكون لكل مناسبة نوع ورد خاص فيها

أو فندم، مثلا أحكم، لون، ملائم، رسم تكون، هناك وتلقائيا نصنر. نقترح الوان، ورد مناسبة

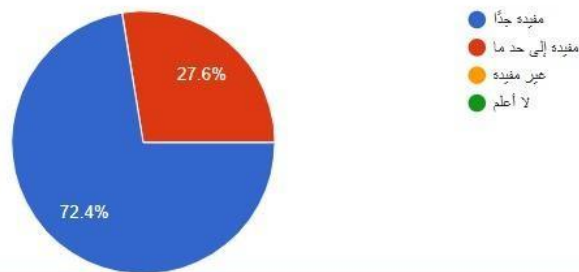
7- ما الميزات التي تجدها مهمة في منصة إلكترونية لشراء الورود؟ (اختر ما تراه مناسباً)

58 ردًا



8- ما رأيك بميزة "المساعد الذكي" التي نقترح بإفادت ورد حسب نوع المناسبة؟

58 ردًا



Conclusions

In a nutshell, this is a well-worth-it attempt at modernizing the flower shopping tradition by developing a special web-based platform for flower shops. The system addresses the typical pains of a customer while ordering flowers: a lack of options to customize, lack of inspiration in choosing the bouquet, and extra time consumption in-store. By developing an intuitive and interactive online environment, the platform allows users to easily navigate through a large variety of flower types, colors, and arrangements, and customize bouquets according to their preferences.

A key contribution of this project involves the integration of the AI Assistant, where users can be supported and assisted through intelligent suggestions about proper bouquet recommendations for occasions like engagements, birthdays, graduations, or other special events. This feature does not only support customers who may lack experience in the art but also saves time through the suggestion of proper flower, color, and arrangement combinations obtained from a dedicated database. Users have every freedom to alter suggested bouquets or create their designs from scratch.

During the development stage, the focus was placed on usability and functionality. The system enables the user to access information regarding the flowers such as name, color, price, and picture, alongside the calculation of the total cost of the flowers ordered by the user. The customer can proceed with the finalization of the order by providing personal details and the desired pickup time of the ordered flowers. The instant confirmation and status messages of the order provide an added facilitative element of communication in the relations of the flower store and the customer.

It also comprises an authentication subsystem that ensures the security of user accounts, alongside the administrative interface for the efficient management of flower stocks, prices, and customer orders. All this supports the flower shop in working efficiently, staying well-organized, and providing improved customer service.

Even with the successful execution of the platform, there had been certain restrictions that had to be overcome. With finite time to develop and the restrictive availability of actual customer feedback in the testing process, there had been certain challenges in the actual evaluation process of the functionality of the system. With the AI-based recommendation service, there had been the limitation of certain pre-developed datasets, thus certain restrictions in personalization in its current state.

On the whole, the designed platform is an example that proves the effectiveness of the combination of web technologies and intelligent systems in improving customer satisfaction, optimizing business processes, and increasing the customer base. It is an ideal foundation for improvements in flower shop services in the future.

Future Work

The proposed project lays an excellent foundation for the development of an intelligent flower shop system. However, if the proposed system is intended to be even more functional and beneficial, the following improvements should be carried out in the future:

Market Expansion:

In

Add extensions to the platform for use by several flower shops or branches in different cities or locations with possibilities of regional integration like language integration, regional flowers, and currency change.

Advanced AI Personalization:

Improve AI Assistant by applying machine learning approaches that evaluate customer behavior, previous orders, and personal preferences for more accurate recommendations of floral arrangements. Online Payment Integration: Integrate secure payment gateways to enable customers to finalize payments online. This should include the ability to pay in different ways such as credit cards, e-wallets, or mobile payment services.

Shipping and Tracking System:

Add the delivery service feature to it, where one can engage in real-time order tracking. That way, the customers can opt for either pickup or delivery and track the status of their orders.

Enhancements in Multimedia and Visualization:

Advanced visualization tools: Make available in the solution 3D bouquet previews or AR features for customers to see what a bouquet will look like when they are ordering it.

Customer Rating and Feedback System:

Provide a system of reviews and ratings of the bouquets and services to get customer feedback in order to continuously improve product quality and user satisfaction.

Programs/Platforms Used

Tool/Platform · Purpose

Figma / Canva UI design and visual prototyping

Adobe Photoshop - Image editing of flower photos

JavaScript Front-end interactivity and logic

HTML & CSS Website structure and styling

Node.js / PHP / ASP.NET Development backend (optional) MySQL / Firebase Database management

GitHub - Version control and collaboration Project Timeline

This included the requirement analysis phase, system design phase, implementation, testing, and documentation within the project timeline. In fact, each phase was completed within the time allocated to that phase, hence ensuring steady progress towards the final delivery of the system.

Code Snippet

- Language switch system (EN / AR)

```
function applyLanguage(lang) {

    document.documentElement.lang = lang;
    document.documentElement.dir = lang === "ar" ? "rtl" : "ltr";

    langBtn.textContent = lang === "en" ? "AR" : "EN";

    allKeys.forEach(el => {
        const key = el.getAttribute("data-key");
        if (translations[lang] && translations[lang][key]) {
            el.textContent = translations[lang][key];
        }
    });

    localStorage.setItem('bf_lang', lang);
    currentLang = lang;
}
```

- Dynamic animated background

```
body::before {
    content: '';
    position: fixed;
    top: 0;
    left: 0;
    width: 100%;
    height: 100%;
    background: radial-gradient(circle at 20% 50%, rgba(184, 76, 101, 0.1), transparent 50%),
                radial-gradient(circle at 80% 20%, rgba(138, 48, 70, 0.1), transparent 50%),
                radial-gradient(circle at 40% 80%, rgba(255, 230, 235, 0.1), transparent 50%);
    animation: morphWaves 15s ease-in-out infinite;
    z-index: -1;
}
```

- Smart Package Suggestion System

```

const occasionSelect = document.getElementById('occasionSelect');
const occasionInput = document.getElementById('occasionInput');
const toneSelect = document.getElementById('toneSelect');
const suggestBtn = document.getElementById('suggestBtn');
const suggestRandomBtn = document.getElementById('suggestRandomBtn');
const suggestionArea = document.getElementById('suggestionArea');
const quantityInput = document.getElementById('quantity');
const deliveryDateInput = document.getElementById('deliveryDate');
const colorChips = document.querySelectorAll('.chip');

let selectedColors = new Set();

colorChips.forEach(chip=>{
  chip.addEventListener('click', ()=>{
    const c = chip.dataset.color;
    if(selectedColors.has(c)){ selectedColors.delete(c); chip.classList.remove('selected');
  }
    else { selectedColors.add(c); chip.classList.add('selected'); }
  });
});

function findBestSuggestion({occasionText='', tone='', colors=[]}) {
  const q = (occasionText || '').toLowerCase().trim();
  const suggestions = DEFAULT_SUGGESTIONS;
  if(suggestions.length === 0) return null;

  const exact = suggestions.find(s=> Array.isArray(s.occasion) && s.occasion.some(o=>
o.toLowerCase() === q ));
  if(exact) return exact;

  for(const s of suggestions){
    if(!s.keywords) continue;
    for(const kw of s.keywords) if(q.includes(kw.toLowerCase())) return s;
  }

  if(tone){
    const byTone = suggestions.find(s=> (s.tone||'').toLowerCase() === tone.toLowerCase());
    if(byTone) return byTone;
  }

  if(colors && colors.length){
    const colorPref = colors.join(',').toLowerCase();
    if(colorPref.includes('red') || colorPref.includes('pink')){
      const byColor = suggestions.find(s => (s.tone||'').toLowerCase() === 'romantic' );
      if(byColor) return byColor;
    }
  }
}

```

```

    }
  }

  const tokens = q.split(/[\\s\\,\\-\\_\\.]+/).filter(Boolean);
  if(tokens.length){
    for(const t of tokens){
      const byTok = suggestions.find(s=> (s.keywords||[]).some(k=>
k.toLowerCase().includes(t)) );
      if(byTok) return byTok;
    }
  }

  return suggestions[Math.floor(Math.random()*suggestions.length)];
}

function renderSuggestion(s){
  if(!s){ suggestionArea.innerHTML = '<p style="color:#777">No suggestions available.</p>';
return; }
  const quantity = Number(quantityInput.value) || 1;
  const delivery = deliveryDateInput.value || 'No date specified';
  const colors = Array.from(selectedColors).join(', ') || 'No preference';
  suggestionArea.innerHTML = `
    
    <div style="flex:1">
      <h4 style="color:#8a3046;margin-bottom:6px">${s.name}</h4>
      <p style="margin:0 0 6px">${s.desc}</p>
      <p style="color:#777;margin:0 0 6px">Delivery: ${delivery} – Quantity: ${quantity} –
Colors: ${colors}</p>
      <div style="display:flex;gap:8px;margin-top:6px">
        <span style="font-weight:800;color:#b84c65">${s.price ? s.price + '₹' : ''}</span>
        <a class="btn" href="${s.link}" style="background:#fff;color:#b84c65;border:1px
solid #b84c65">View product</a>
        <button class="btn" id="addToCartBtn">Add to cart</button>
      </div>
    </div>
  `;
  suggestionArea.appendChild(div);
  document.getElementById('addToCartBtn').addEventListener('click', ()=>{
    let cart = JSON.parse(localStorage.getItem('bf_cart_demo') || '[]');
    cart.push({id:s.id, name:s.name, price:s.price, qty:quantity, delivery:delivery,
colors:Array.from(selectedColors)});
    localStorage.setItem('bf_cart_demo', JSON.stringify(cart));
    alert(s.name + ' added to demo cart. Go to home and open cart to view (demo).');
  });
}

```

```

    });
  }

  suggestBtn.addEventListener('click', ()=>{
    const sel = occasionSelect.value || '';
    const custom = occasionInput.value.trim();
    const q = custom || sel;
    const tone = toneSelect.value || '';
    if(!q){ suggestionArea.innerHTML = '<p style="color:#777">Please choose an occasion or
type it in the field.</p>'; return; }
    const best = findBestSuggestion({occasionText:q, tone,
colors:Array.from(selectedColors)});
    renderSuggestion(best);
  });

  suggestRandomBtn.addEventListener('click', ()=>{
    const rand = DEFAULT_SUGGESTIONS[Math.floor(Math.random()*DEFAULT_SUGGESTIONS.length)];
    renderSuggestion(rand);
  });

  suggestionArea.innerHTML = '<p style="color:#777">Choose occasion or click "Random
suggestion" to see a bouquet.</p>';
</script>

```

- Page Protection

```

if(localStorage.getItem('bf_logged_in') !== '1'){
  window.location.href = 'login.html';
}

```

- Smart Addition Logic

```

const uid = sp.id + '::' + (colorKey || 'mixed');
if(bouquet[uid])
  bouquet[uid].qty += qty;
else bouquet[uid] = { uid,
  speciesId: sp.id,
  species: sp.species,
  colorKey: colorKey || 'mixed',
  colorLabel,
  price: sp.price,
  qty, img: sp.img,
  colorHex
};

```

- Automatic price calculation

```
function updateTotals(){
  const sub = Math.round(calcSubtotal()*100)/100;
  const wrapping = sub > 0 ? 5 : 0;
  const tax = Math.round((sub + wrapping) * 0.05 * 100) / 100;
  const total = Math.round((sub + wrapping + tax) * 100) / 100;
  document.getElementById('subtotal').textContent = formatPrice(sub);
  document.getElementById('wrapping').textContent = formatPrice(wrapping);
  document.getElementById('tax').textContent = formatPrice(tax);
  document.getElementById('total').textContent = formatPrice(total);
}
```

- Fully dynamic render

```
function renderSpecies(list){
  flowersGrid.innerHTML = '';
  list.forEach(sp => {
    const card = document.createElement('div');
    card.className = 'flower-card';
    card.innerHTML = `
      <div class="flower-media"></div>
      <div class="flower-info">
        <div class="flower-title">
          <div>
            <div class="name">${sp.species}</div>
            <div class="species-note">${sp.desc}</div>
          </div>
          <div class="price">${formatPrice(sp.price)}</div>
        </div>

        <div style="margin-top:8px">
          <div style="font-
size:13px;color:${getComputedStyle(document.documentElement).getPropertyValue('--
muted')}>Available colors:</div>
          <div class="swatches"></div>
        </div>

        <div class="card-controls">
          <button class="qty-btn" data-action="decrease">-</button>
          <input class="qty-input" type="number" min="0" value="1">
          <button class="qty-btn" data-action="increase">+</button>
          <div style="flex:1"></div>
          <button class="add-btn">Add to bouquet</button>
        </div>
    `;
  });
}
```

```

</div>
`;

const swatchesEl = card.querySelector('.swatches');
let selectedColorKey = '';
sp.colors.forEach(c=>{
  const sw = document.createElement('div');
  sw.className = 'swatch';
  sw.title = c.label;
  sw.style.background = c.hex;
  sw.dataset.key = c.key;
  sw.addEventListener('click', ()=>{
    swatchesEl.querySelectorAll('.swatch').forEach(s=>s.classList.remove('selected'));
    sw.classList.add('selected');
    selectedColorKey = c.key;
  });
  swatchesEl.appendChild(sw);
});

const inc = card.querySelector('[data-action="increase"]');
const dec = card.querySelector('[data-action="decrease"]');
const qtyInput = card.querySelector('.qty-input');
const addBtn = card.querySelector('.add-btn');

inc.addEventListener('click', ()=> qtyInput.value = Math.max(0, Number(qtyInput.value || 0)
+ 1));
dec.addEventListener('click', ()=> qtyInput.value = Math.max(0, Number(qtyInput.value || 0)
- 1));
qtyInput.addEventListener('input', ()=>{ if(qtyInput.value === '' || Number(qtyInput.value)
< 0) qtyInput.value = 0; });

addBtn.addEventListener('click', ()=>{
  const qty = Math.max(0, Math.floor(Number(qtyInput.value) || 0));
  if(qty === 0){ alert('Quantity must be at least 1'); return; }

  let colorKey = selectedColorKey || (sp.colors[0] ? sp.colors[0].key : '');
  let colorLabel = 'Mixed';
  let colorHex = '#ddd';
  const chosen = sp.colors.find(c=>c.key === colorKey);
  if(chosen){ colorLabel = chosen.label; colorHex = chosen.hex; }

  const uid = sp.id + '::' + (colorKey || 'mixed');
  if(bouquet[uid]) bouquet[uid].qty += qty;
  else bouquet[uid] = { uid, speciesId: sp.id, species: sp.species, colorKey: colorKey ||
'mixed', colorLabel, price: sp.price, qty, img: sp.img, colorHex };

```

```

    saveLocal();
    renderBouquet();
    updateTotals();

    addBtn.textContent = 'Added ✓';
    addBtn.disabled = true;
    setTimeout(()=>{ addBtn.textContent = 'Add to bouquet'; addBtn.disabled = false; }, 700);
  });

  flowersGrid.appendChild(card);
});
}

```

```

function renderBouquet(){
  itemsBox.innerHTML = '';
  const keys = Object.keys(bouquet);
  if(keys.length === 0){
    document.getElementById('emptyHint').style.display = 'block';
    updateTotals();
    return;
  }
  document.getElementById('emptyHint').style.display = 'none';
  keys.forEach(k=>{
    const it = bouquet[k];
    const div = document.createElement('div');
    div.className = 'item';
    div.innerHTML = `
      <div class="item-img"></div>
      <div class="item-body">
        <div style="font-weight:800">${it.species}</div>
        <div class="muted">${it.colorLabel} • ${formatPrice(it.price)} each</div>
      </div>
      <div style="display:flex;flex-direction:column;align-items:flex-end;gap:8px">
        <div class="item-qty">
          <button class="qty-btn" data-action="dec">-</button>
          <input type="number" min="0" value="${it.qty}" style="width:56px;padding:6px;border-radius:8px;border:1px solid rgba(0,0,0,0.06);text-align:center">
          <button class="qty-btn" data-action="inc">+</button>
        </div>
        <div class="item-subtotal">${formatPrice(it.price * it.qty)}</div>
      </div>
    `;
    itemsBox.appendChild(div);
  });
  updateTotals();
}

```

```

        <button class="remove-btn" style="background:transparent;border:none;color:var(--muted);cursor:pointer">Remove</button>
      </div>
    `;

    const dec = div.querySelector('[data-action="dec"]');
    const inc = div.querySelector('[data-action="inc"]');
    const qtyIn = div.querySelector('input[type="number"]');
    const rm = div.querySelector('.remove-btn');

    dec.addEventListener('click', ()=>{ let v=Math.max(0,Number(qtyIn.value)||0)-1;
    qtyIn.value=v; updateQty(k,v); });
    inc.addEventListener('click', ()=>{ let v=Math.max(0,Number(qtyIn.value)||0)+1;
    qtyIn.value=v; updateQty(k,v); });
    qtyIn.addEventListener('input', ()=>{ let v=Math.max(0,Math.floor(Number(qtyIn.value)||0));
    qtyIn.value=v; updateQty(k,v); });
    rm.addEventListener('click', ()=>{ delete bouquet[k]; saveLocal(); renderBouquet();
    updateTotals(); });

    itemsBox.appendChild(div);
  });
}

```

- **Shopping Cart Logic**

```
let cart = loadCart();
```

Add Product:

```

function addToCart(id){
  const prod = PRODUCTS.find(p=>p.id===id);
  if(!prod) return;
  const existing = cart.find(i=>i.id===id);
  if(existing) existing.qty += 1;
  else cart.push({ id: prod.id, title: prod.title, price: prod.price, img: prod.img, qty: 1
});
  saveCart();
  renderCart();
}

```

View Cart :

```

function renderCart(){
  cartListEl.innerHTML = '';

```



```

if(cart.length===0){
  cartListEl.innerHTML = '<div class="empty">السلة فارغة — اصف منتجات بالضغط على "Order Now"</div>';
  cartTotalEl.textContent = '0.00 $';
  cartCountEl.textContent = '0';
  return;
}
cart.forEach(item=>{
  const div = document.createElement('div');
  div.className = 'cart-item';
  div.innerHTML = `
    
    <div class="meta">
      <div style="font-weight:600">${item.title}</div>
      <div style="color:var(--muted);font-size:0.95rem">${item.price.toFixed(2)} $</div>
    </div>
    <div class="qty-controls">
      <button class="btn ghost" data-action="dec" data-id="${item.id}">-</button>
      <div>${item.qty}</div>
      <button class="btn ghost" data-action="inc" data-id="${item.id}">+</button>
      <button class="btn" style="margin-left:6px" data-action="remove" data-
id="${item.id}">حذف</button>
    </div>
  `;
  cartListEl.appendChild(div);
});

cartListEl.querySelectorAll('[data-action]').forEach(b=>{
  b.addEventListener('click', e=>{
    const id = b.dataset.id;
    const action = b.dataset.action;
    if(action==='inc') changeQty(id, +1);
    if(action==='dec') changeQty(id, -1);
    if(action==='remove') removeItem(id);
  });
});

const total = cart.reduce((s,i)=>s + i.price * i.qty, 0);
cartTotalEl.textContent = total.toFixed(2) + ' $';
cartCountEl.textContent = cart.reduce((s,i)=>s+i.qty,0);
}

```

Quantity control (+ / -) with automatic deletion:

```

function changeQty(id, delta){
  const it = cart.find(i=>i.id===id);
  if(!it) return;

```

```

    it.qty += delta;
    if(it.qty <= 0) removeItem(id);
    saveCart(); renderCart();
  }

  function removeItem(id){
    cart = cart.filter(i=>i.id!==id);
    saveCart(); renderCart();
  }

  function clearCart(){ cart = []; saveCart(); renderCart(); }

  function checkout(){
    if(cart.length===0){ alert('السلة فارغة'); return; }
    const summary = cart.map(i=>`${i.title} x${i.qty}`).join('\n');
    alert('إرسال الطلب (تجريبي) \n\n' + summary);
    clearCart();
  }

```

Local storage:

```

function saveCart(){ localStorage.setItem(CART_KEY, JSON.stringify(cart)); }
function loadCart(){ try{ return JSON.parse(localStorage.getItem(CART_KEY)) || []; }
} catch(e){ return []; } }

document.getElementById('view-cart').addEventListener('click', ()=>{
  productsView.style.display = 'none';
  cartView.style.display = 'block';
});
document.getElementById('view-products').addEventListener('click', ()=>{
  productsView.style.display = 'block';
  cartView.style.display = 'none';
});

```

• Signup Logic

```

const signupForm = document.querySelector('.signup-form');
signupForm.addEventListener('submit', async (e) => {
  e.preventDefault();
  const name = document.getElementById('name').value;
  const email = document.getElementById('email').value;
  const password = document.getElementById('password').value;
  const confirmPassword = document.getElementById('confirm-password').value;

  if (password !== confirmPassword) return alert("Passwords do not match");

  const res = await fetch('http://localhost:5000/api/auth/register', {

```

```
method: 'POST',
headers: { 'Content-Type': 'application/json' },
body: JSON.stringify({ name, email, password })
});

const data = await res.json();
if (res.ok) {
  alert('Account created successfully!');
  window.location.href = 'login.html';
} else alert(data.message);
});
```